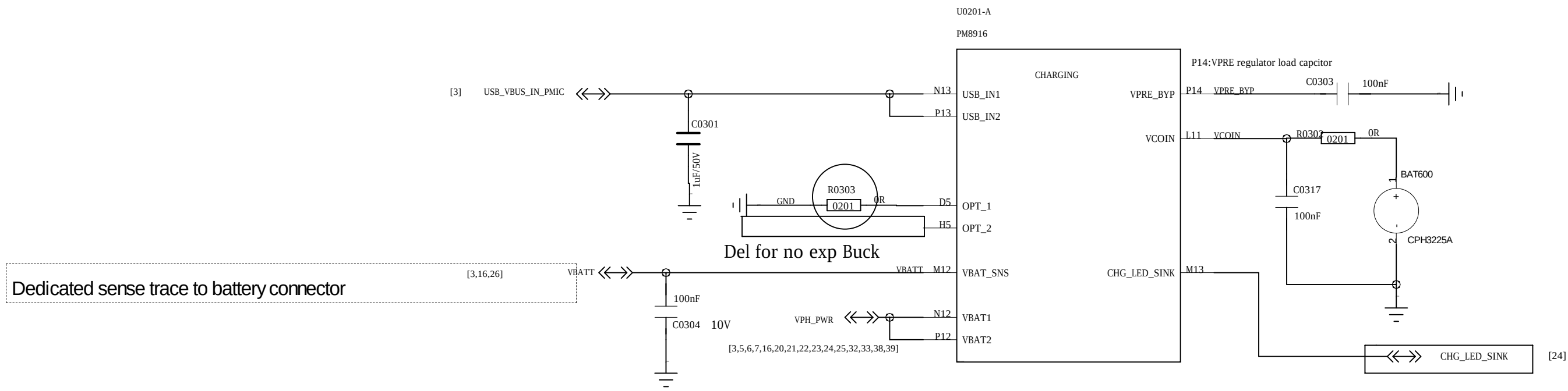
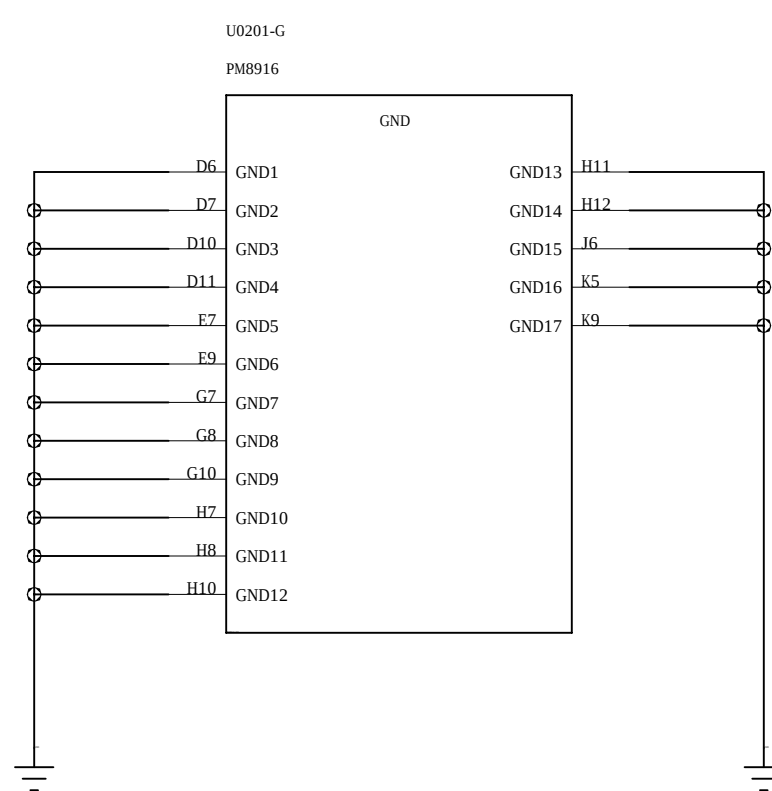
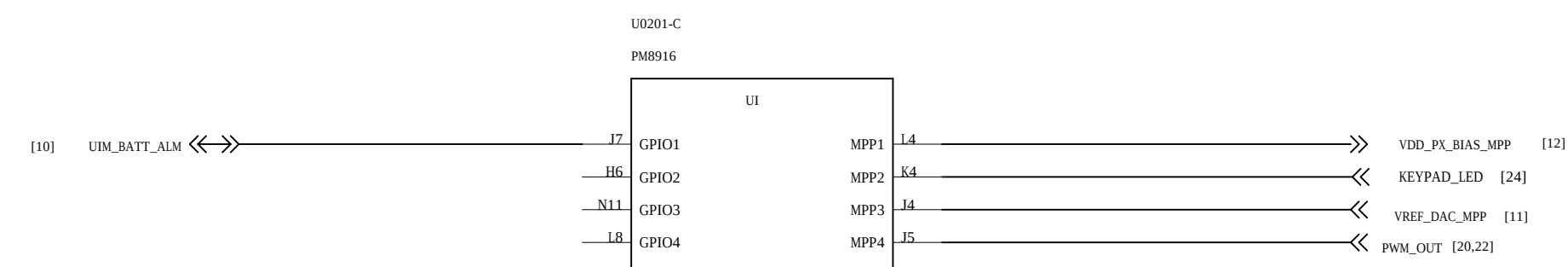


Sheet #		Content			Sheet #		Content		
01.		Table of Content							
02.		PM8916 Control and MPP/Clock							
03.		PM8916 Charging							
04.		PM8916 GPIO/MPP							
05.		PM8916 Buck converter							
06.		PM8916 LDO							
07.		PM8916 Codec							
08.		MSM8916 Control							
09.		MSM8916 EBI							
10.		MSM8916 GPIO							
11.		MSM8916 MIPI and RF Control							
12.		MSM8916 Power1							
13.		MSM8916 Power2							
14.		MSM8916 GND and NC							
15.		MEMORY							
16.		Battery/USB							
17.		Speaker and Vibrator							
18.		MIC and Receiver							
19.		Earphone							
20.		LCD							
21.		Camera							
22.		Sensor							
23.		UIM/TF card							
24.		Keypad/LED							
25.		Touch interface							
26.		Test Points/Shields							
27.		QCA1990 NFC							
28.		Reserved							
29.		Reserved							
30.		Reserved							

Pins	Hi Z	GND
OPT_1	No SMB1360	SMB1360 present
OPT_2	No Ext APC buck (S5)	Ext APC buck (S5) present





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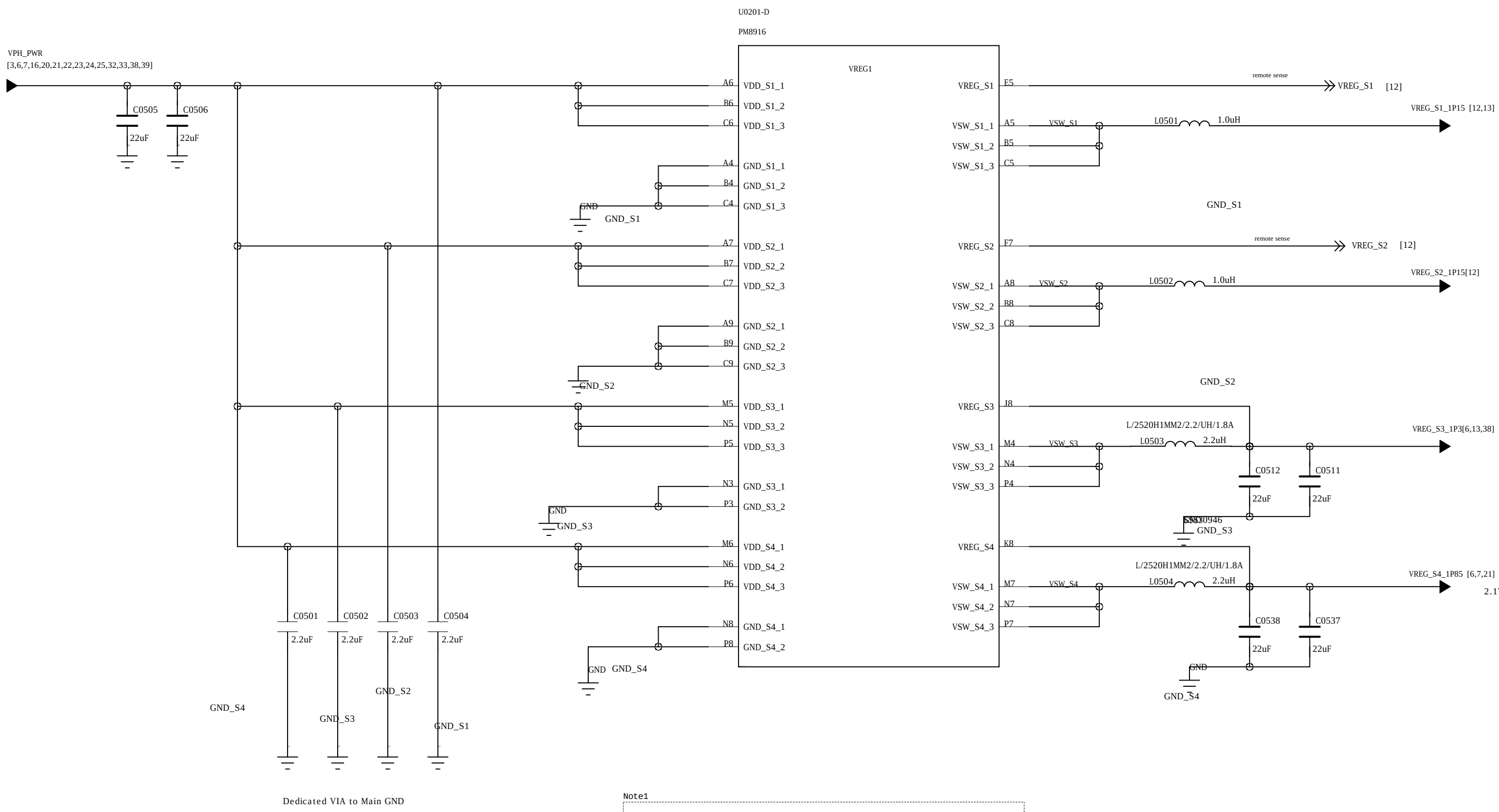
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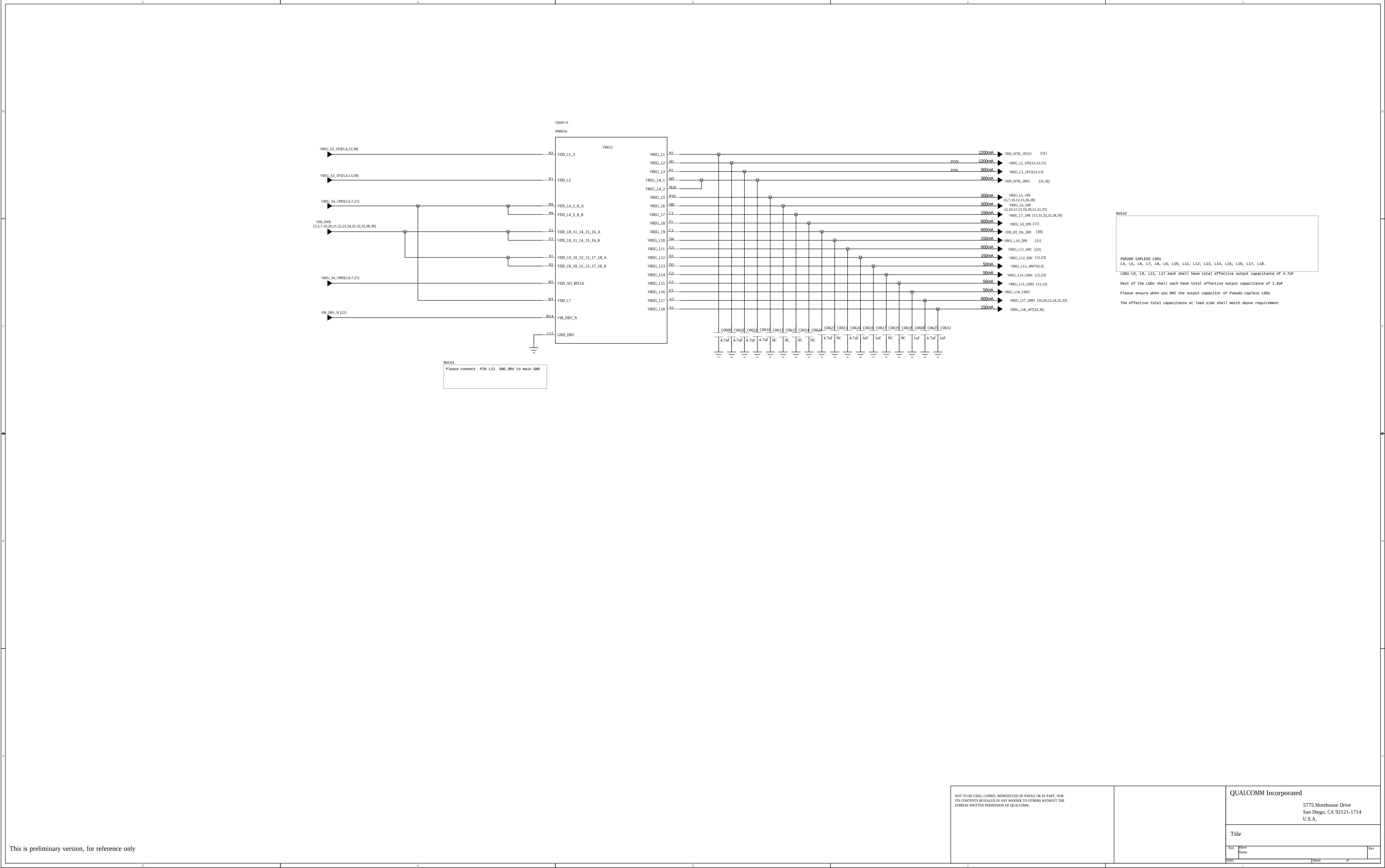
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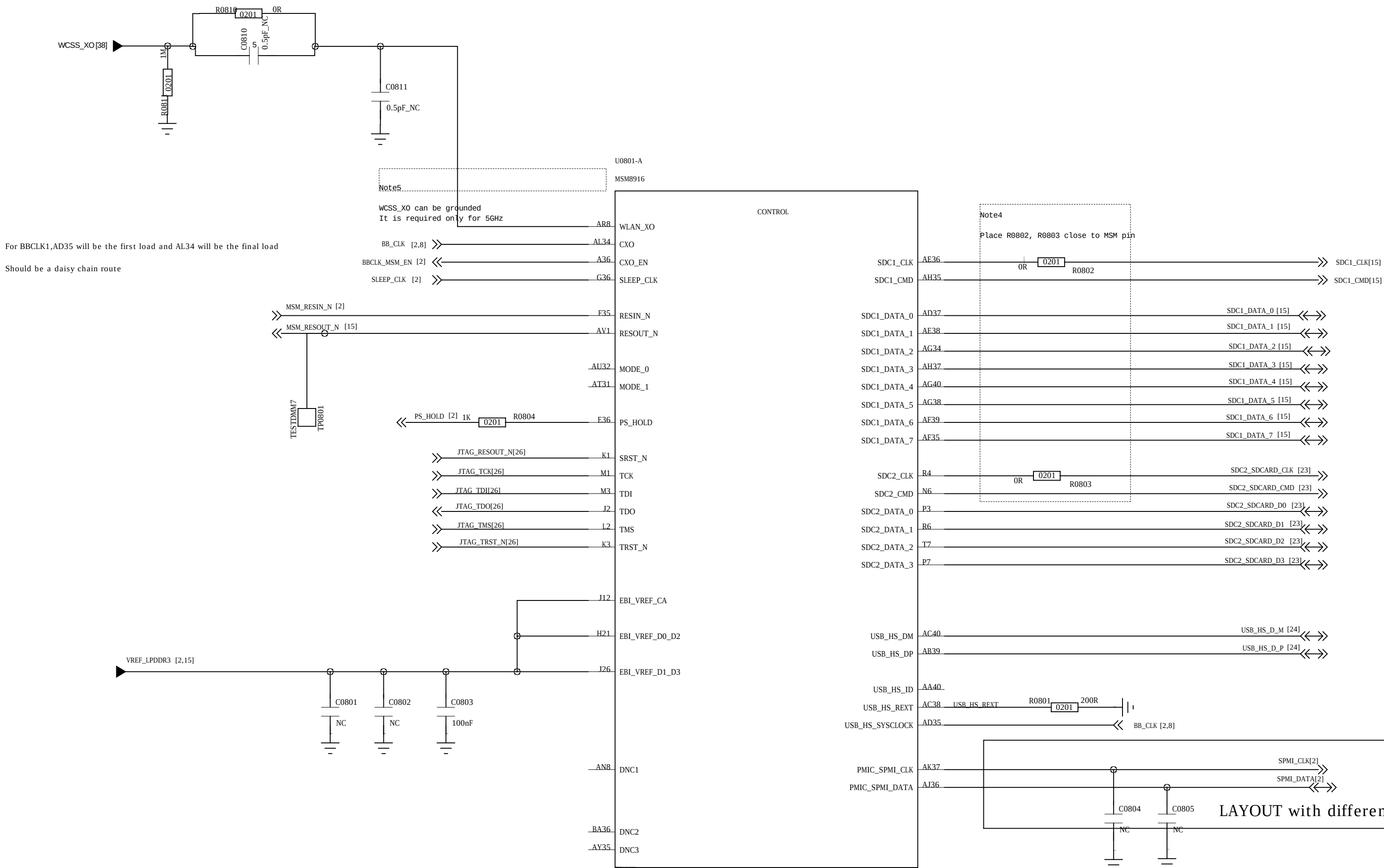
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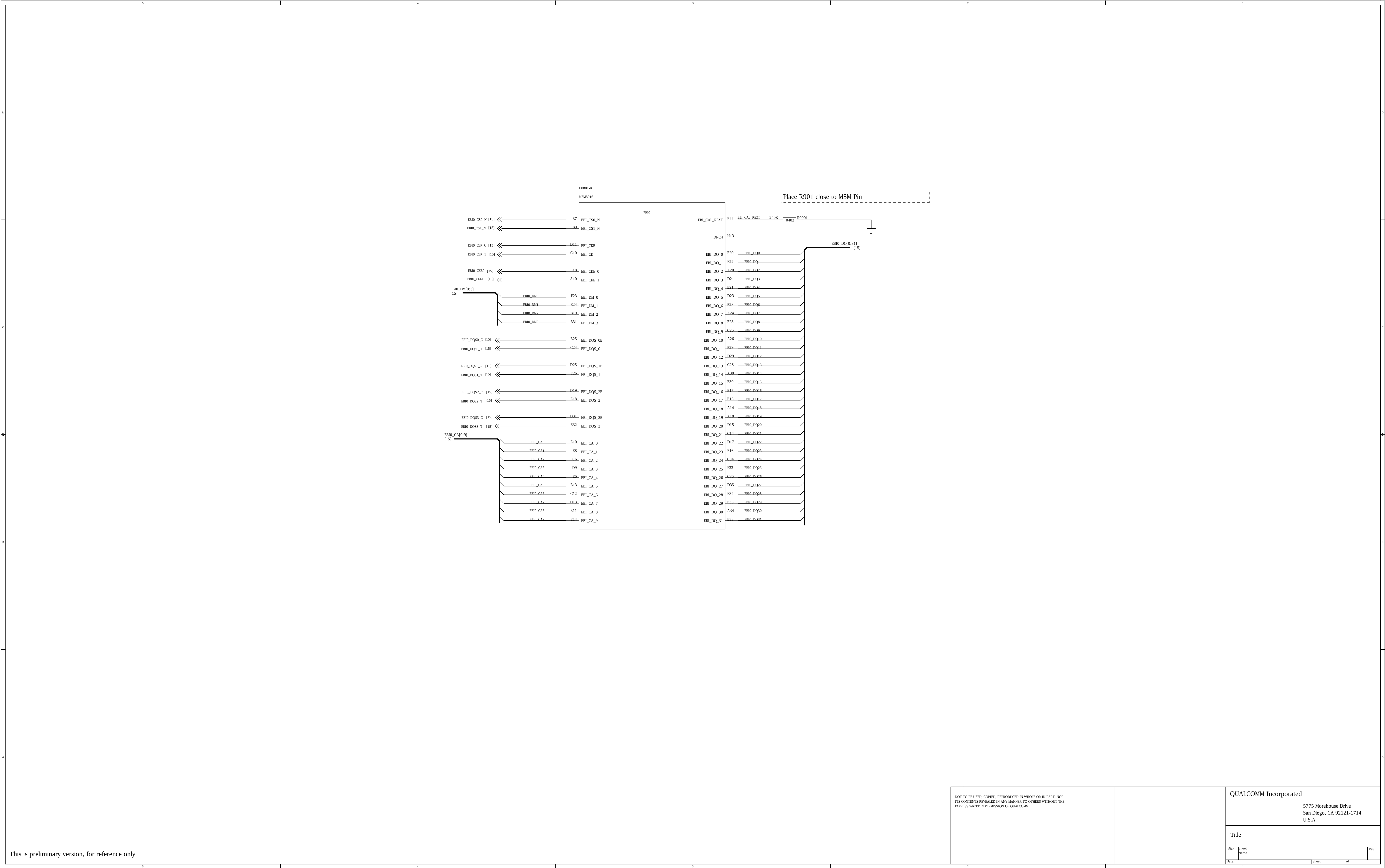


For BRCLK1_AD35 will be the first load and AL34 will be the final load
Should be a daisy chain route

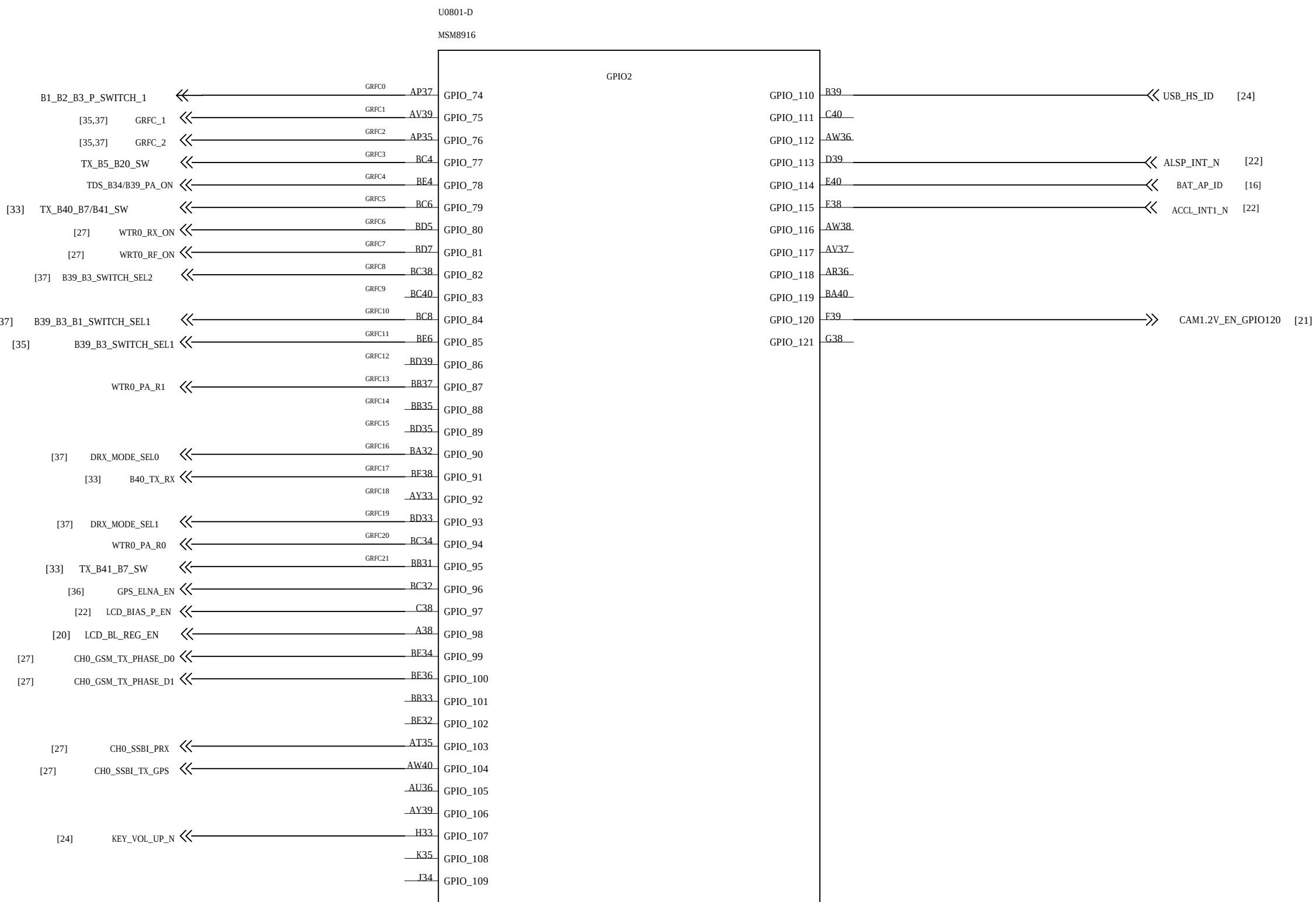
Note4
Place R0802, R0803 close to MSM pin

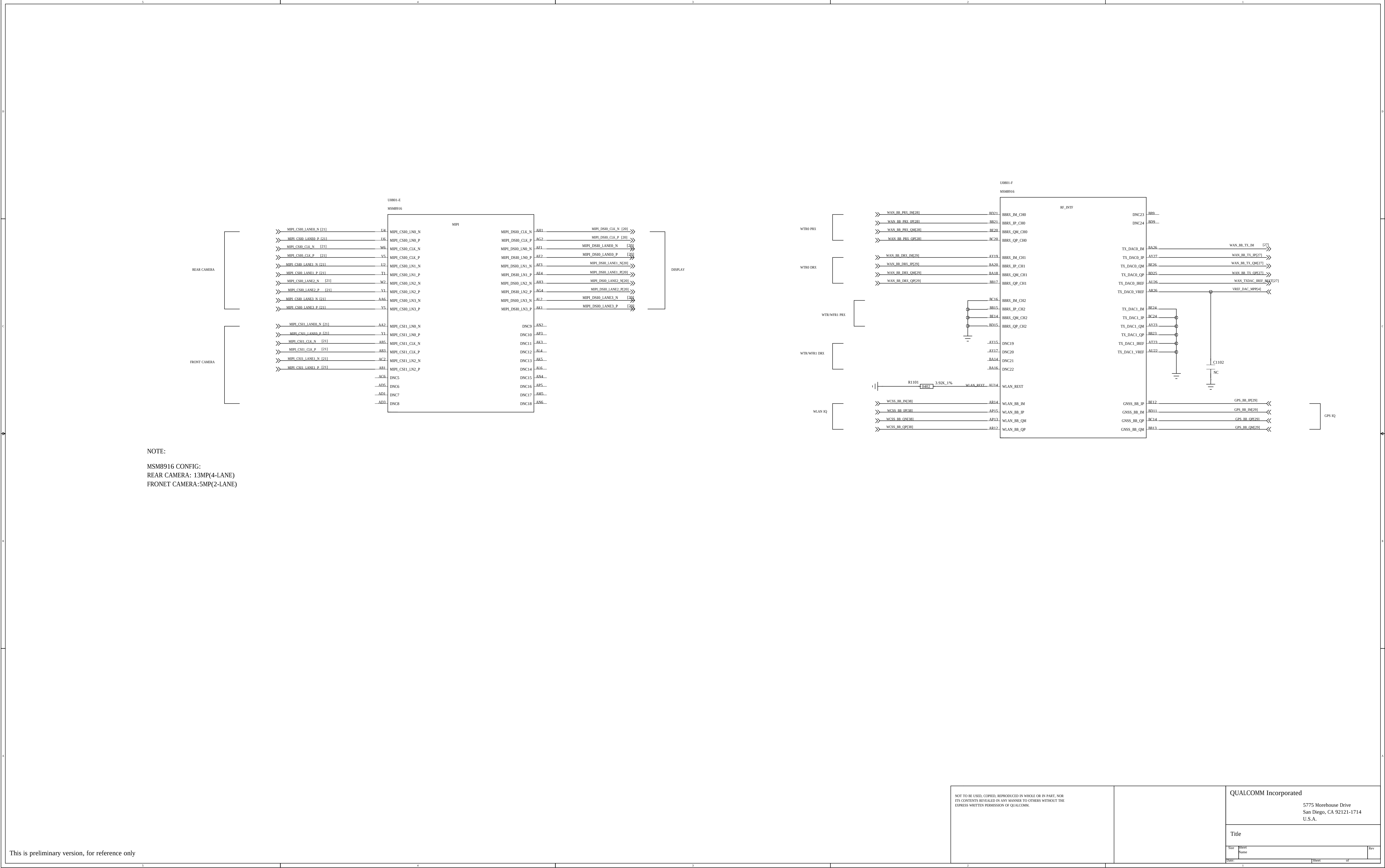
LAYOUT with differential pair

caps must be no more than 2mm away from the MSM



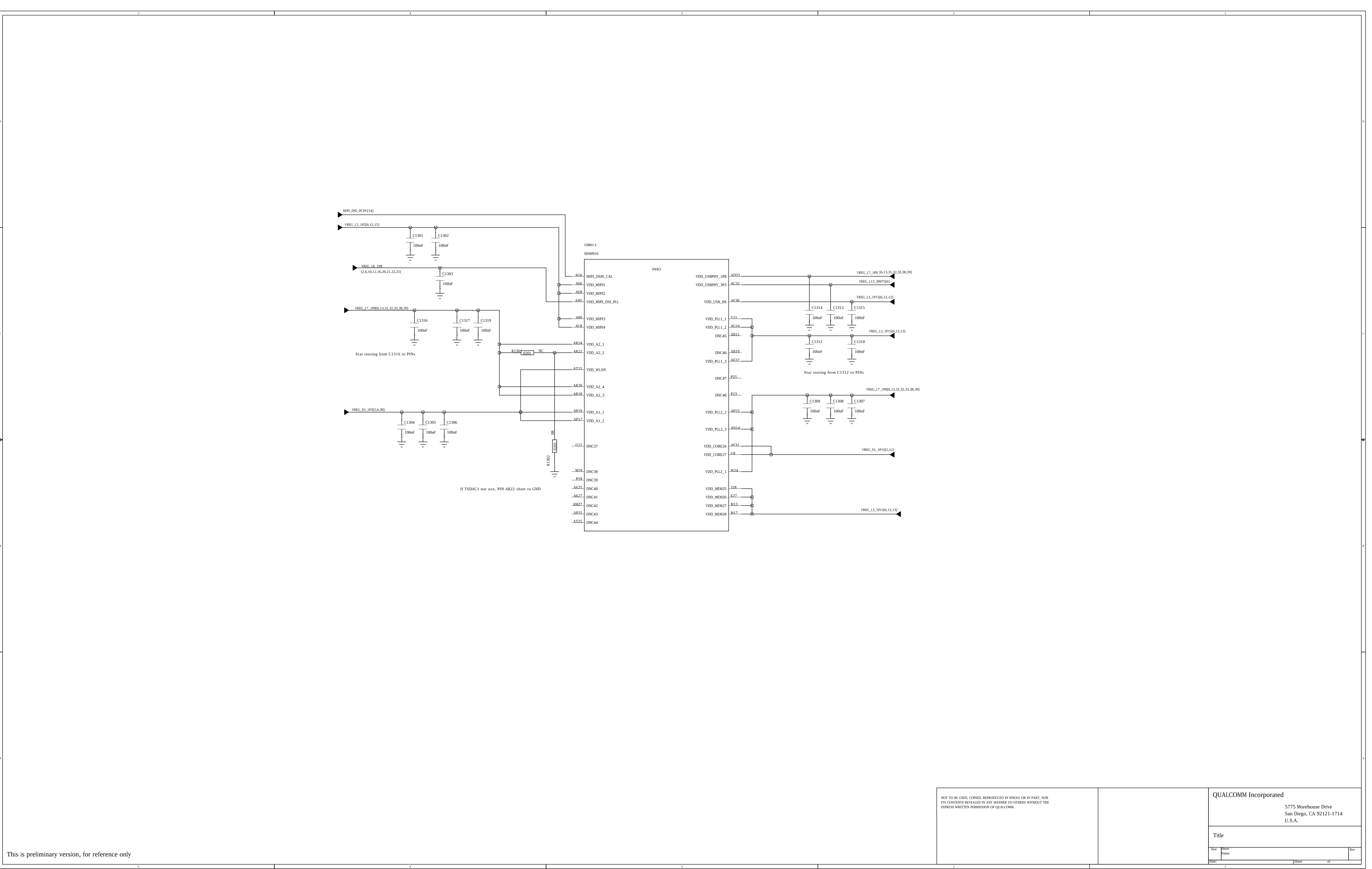
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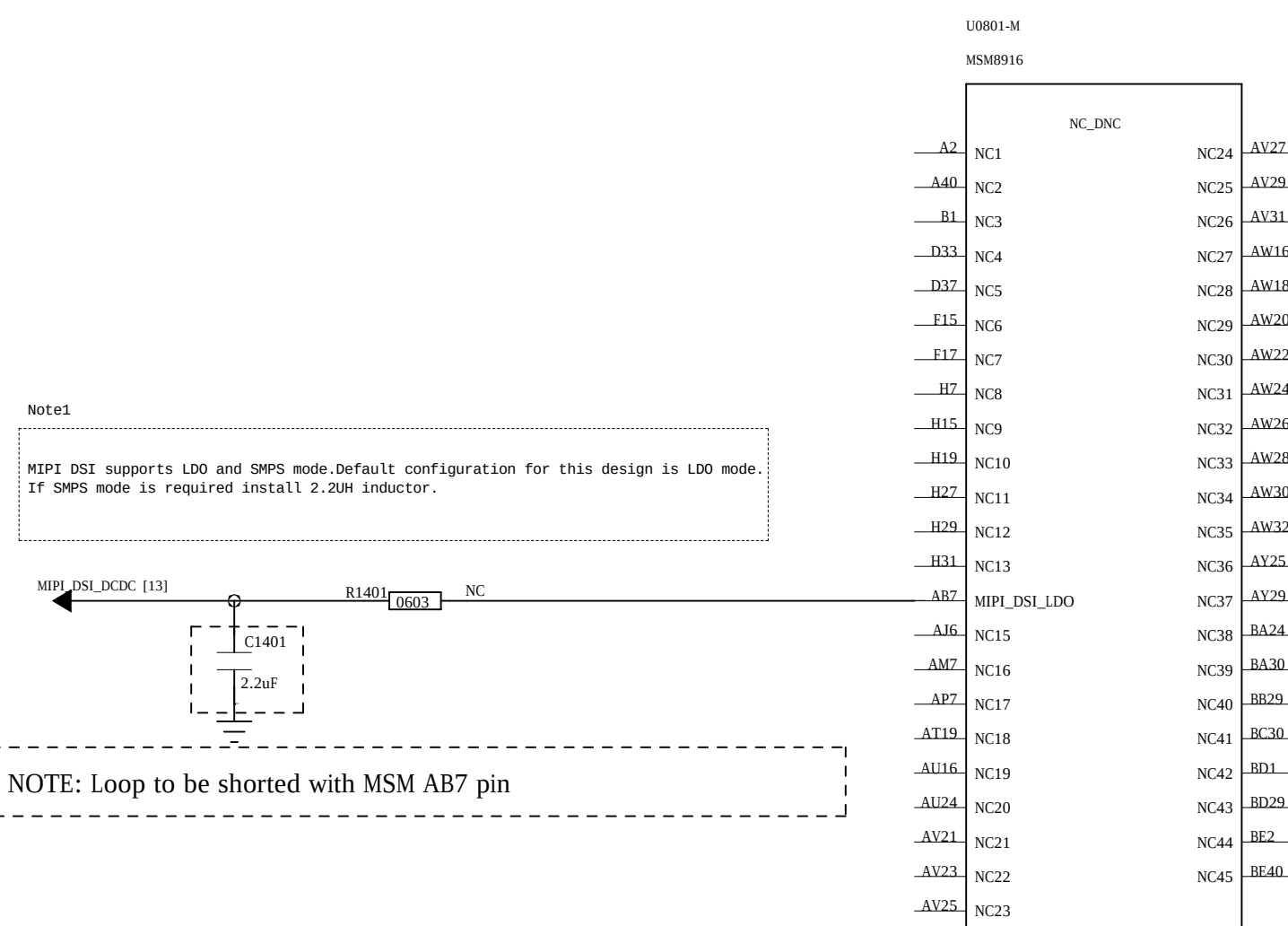
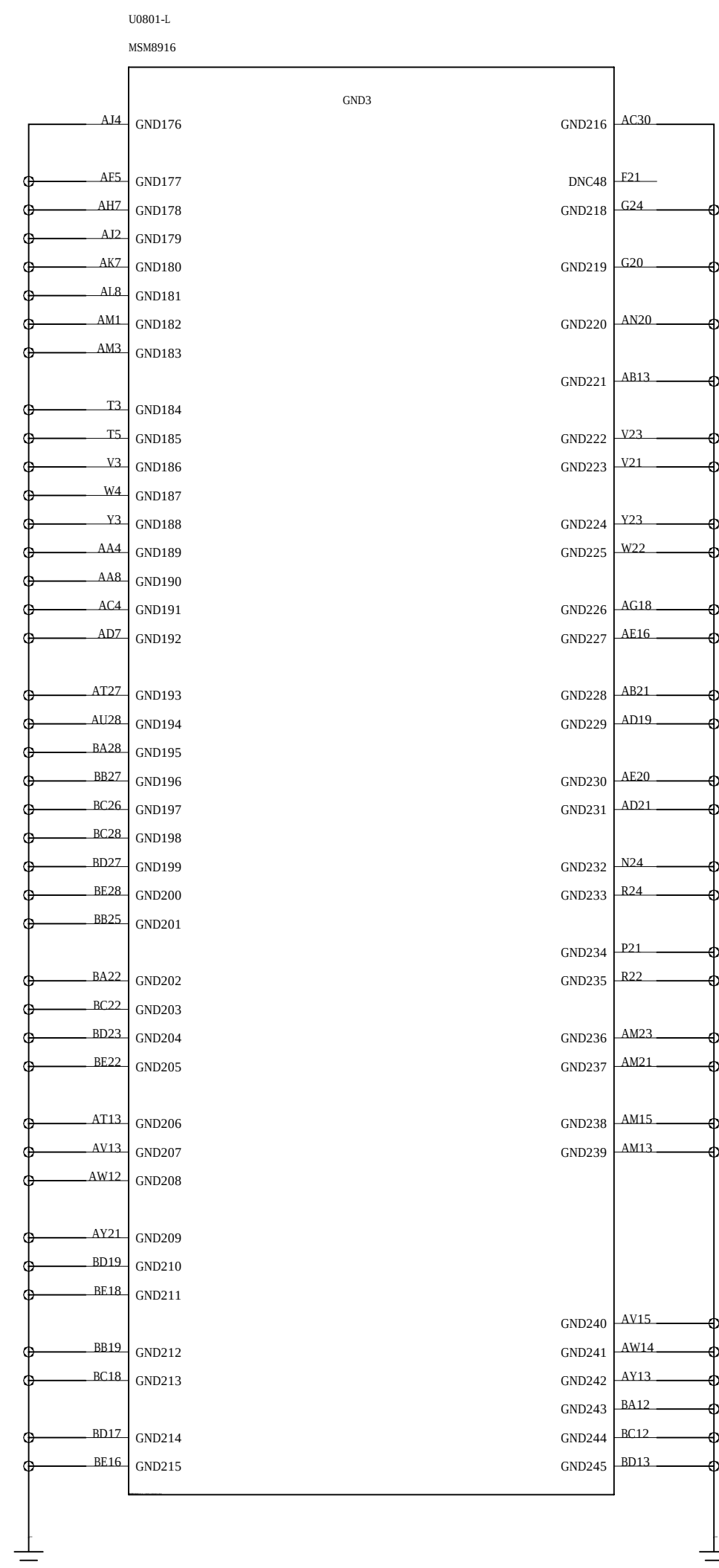
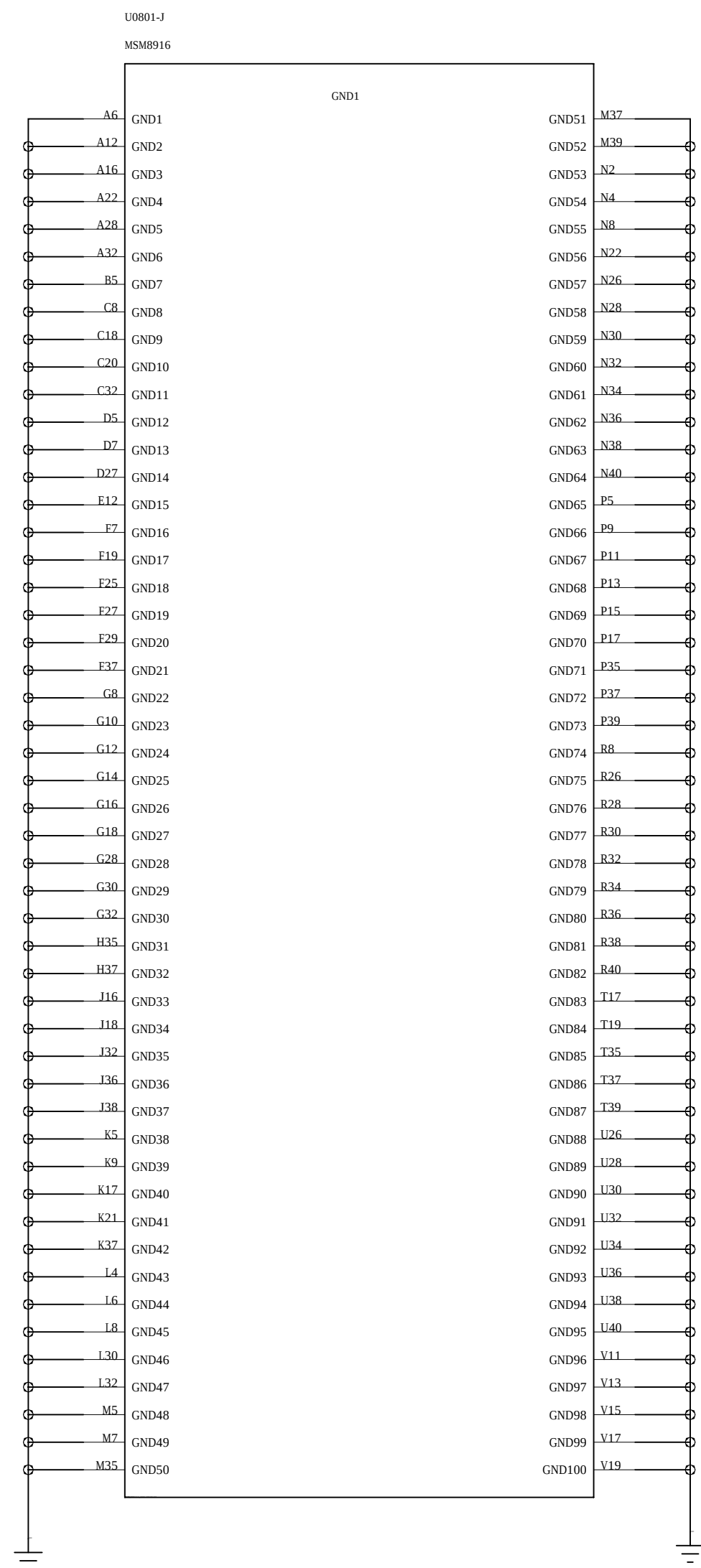
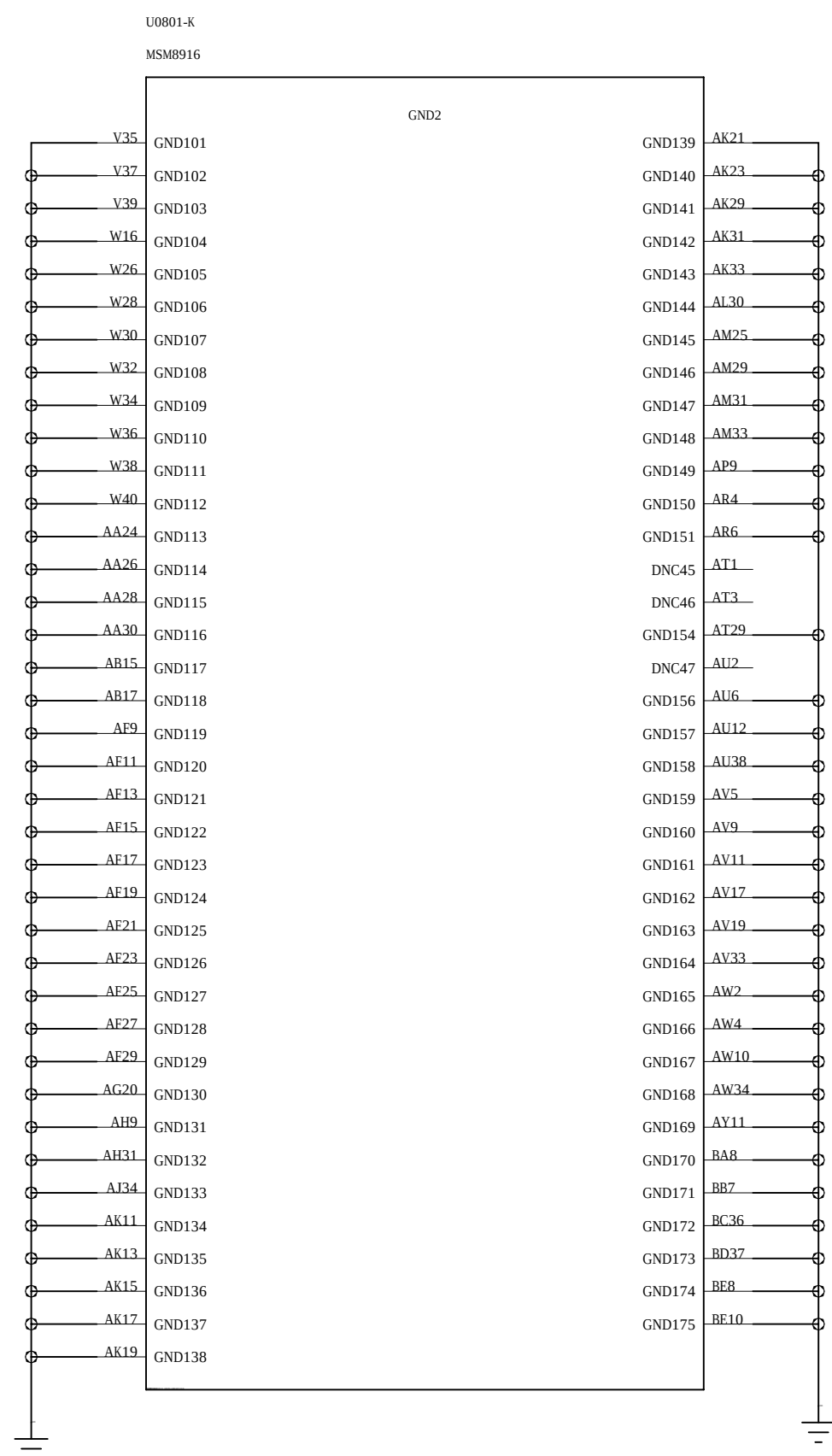


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		Title			
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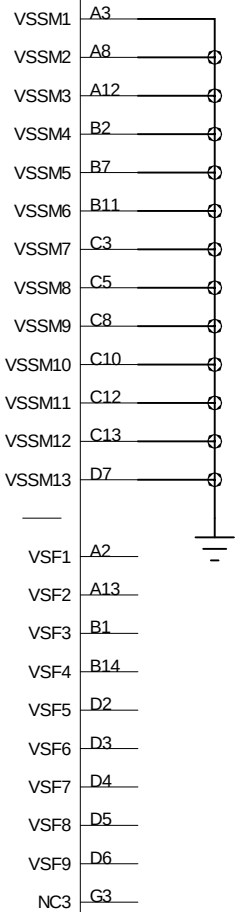
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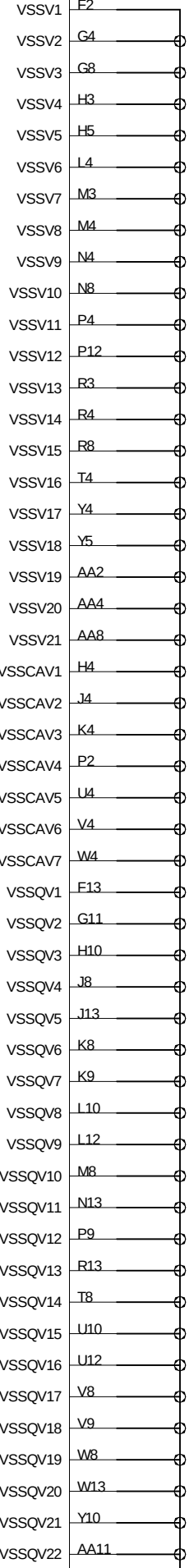
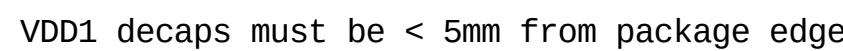
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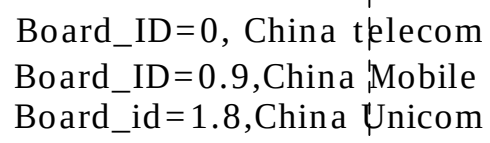
NAME	ADDRESS	DATA	DESCRIPTION
DAT7M	B4	8	SOCl_DATA_7 [8]
DAT6M	A5	8	SOCl_DATA_6 [8]
DAT5M	A10	8	SOCl_DATA_5 [8]
DAT4M	C9	8	SOCl_DATA_4 [8]
DAT3M	B5	8	SOCl_DATA_3 [8]
DAT2M	C6	8	SOCl_DATA_2 [8]
DAT1M	B10	8	SOCl_DATA_1 [8]
DAT0M	A9	8	SOCl_DATA_0 [8]

PC31V	F12	EB0_pg31	EB0_DX[0:31] [8]
PC30V	F11	EB0_pg30	
PC29V	F10	EB0_pg29	
PC28V	G13	EB0_pg28	
PC27V	G12	EB0_pg27	
PC26V	G10	EB0_pg26	
PC25V	H13	EB0_pg25	
PC24V	H11	EB0_pg24	
PC23V	V11	EB0_pg23	
PC22V	V13	EB0_pg22	
PC21V	AA10	EB0_pg21	
PC20V	AA12	EB0_pg20	
PC19V	AA13	EB0_pg19	
QC18V	AB10	EB0_pg18	
QC17V	AB11	EB0_pg17	
QC16V	AB12	EB0_pg16	
QC15V	J12	EB0_pg15	
QC14V	K13	EB0_pg14	
QC13V	K11	EB0_pg13	
QC12V	L11	EB0_pg12	
QC11V	L13	EB0_pg11	
QC10V	M11	EB0_pg10	
QC9V	M13	EB0_pg9	
QC8V	N12	EB0_pg8	
QC7V	R12	EB0_pg7	
QC6V	T13	EB0_pg6	
QC5V	T11	EB0_pg5	
QC4V	U13	EB0_pg4	
QC3V	U11	EB0_pg3	
QC2V	V13	EB0_pg2	
QC1V	V11	EB0_pg1	
QC0V	W12	EB0_pg0	



VDD2 decaps must be $< 1.5\text{mm}$ from package edge

Board_ID=0, China telecom
Board_ID=0.9, China Mobile
Board_id=1.8, China Unicom



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Title	Author	Year	Journal	Volume	Page
1. The Effect of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	1-15
2. The Impact of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	16-30
3. The Effect of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	31-45
4. The Impact of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	46-60
5. The Effect of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	61-75
6. The Impact of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	76-90
7. The Effect of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	91-105
8. The Impact of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	106-120
9. The Effect of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	121-135
10. The Impact of the 1997 Asian Financial Crisis on the U.S. Economy	John H. Coatsworth	1998	Journal of International Economics	50	136-150

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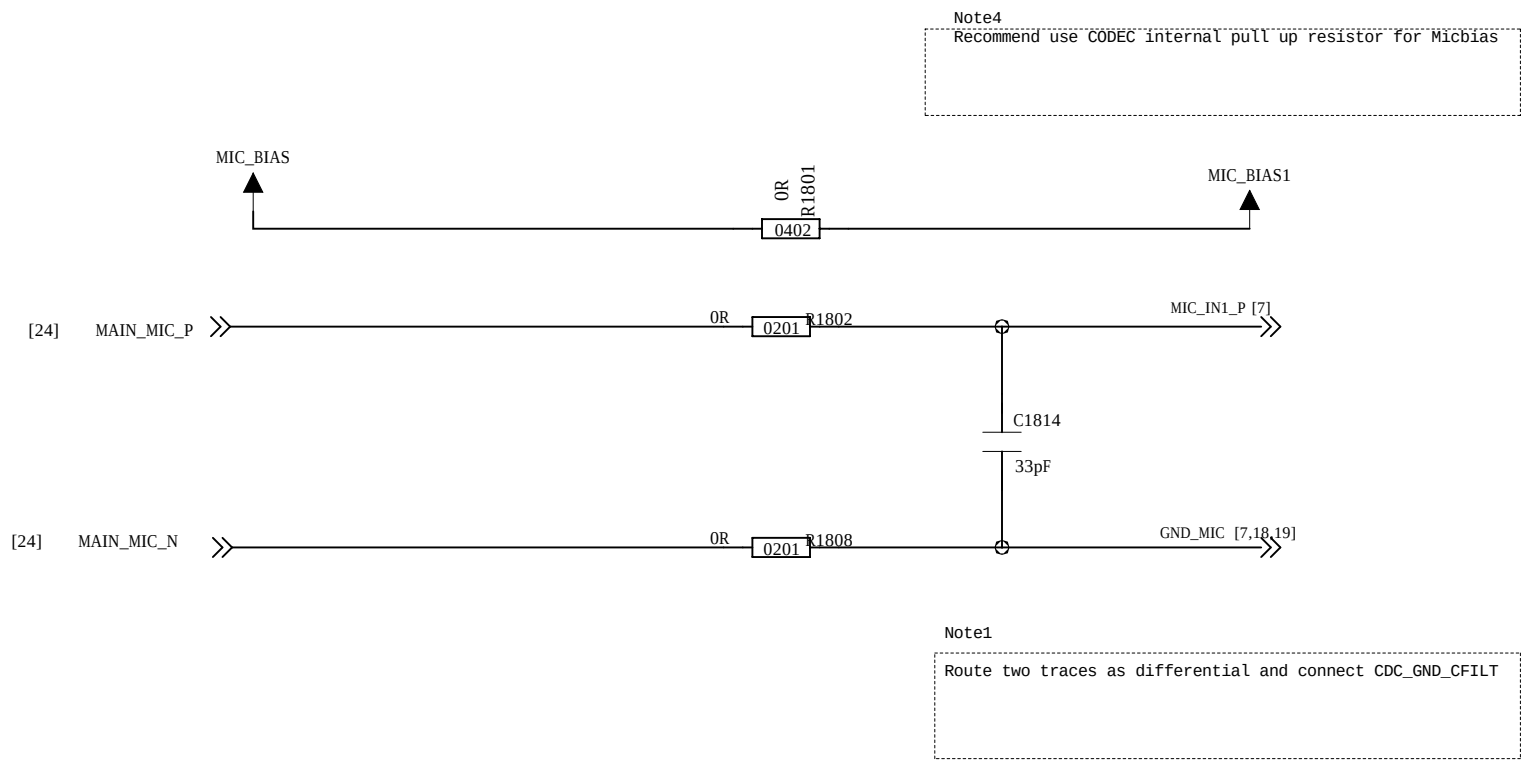
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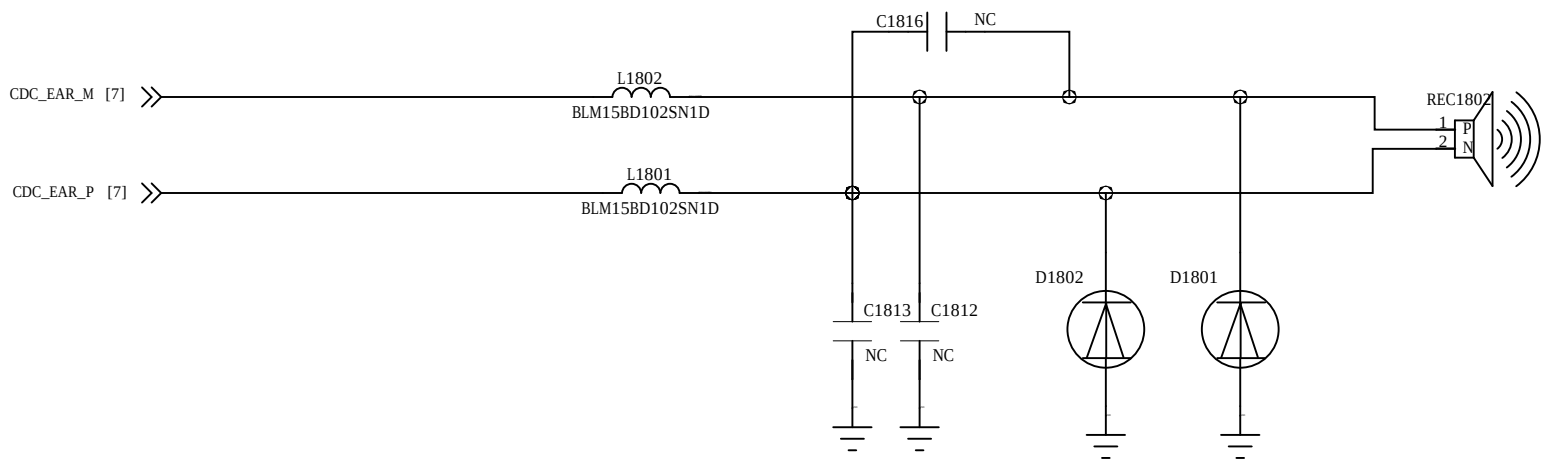
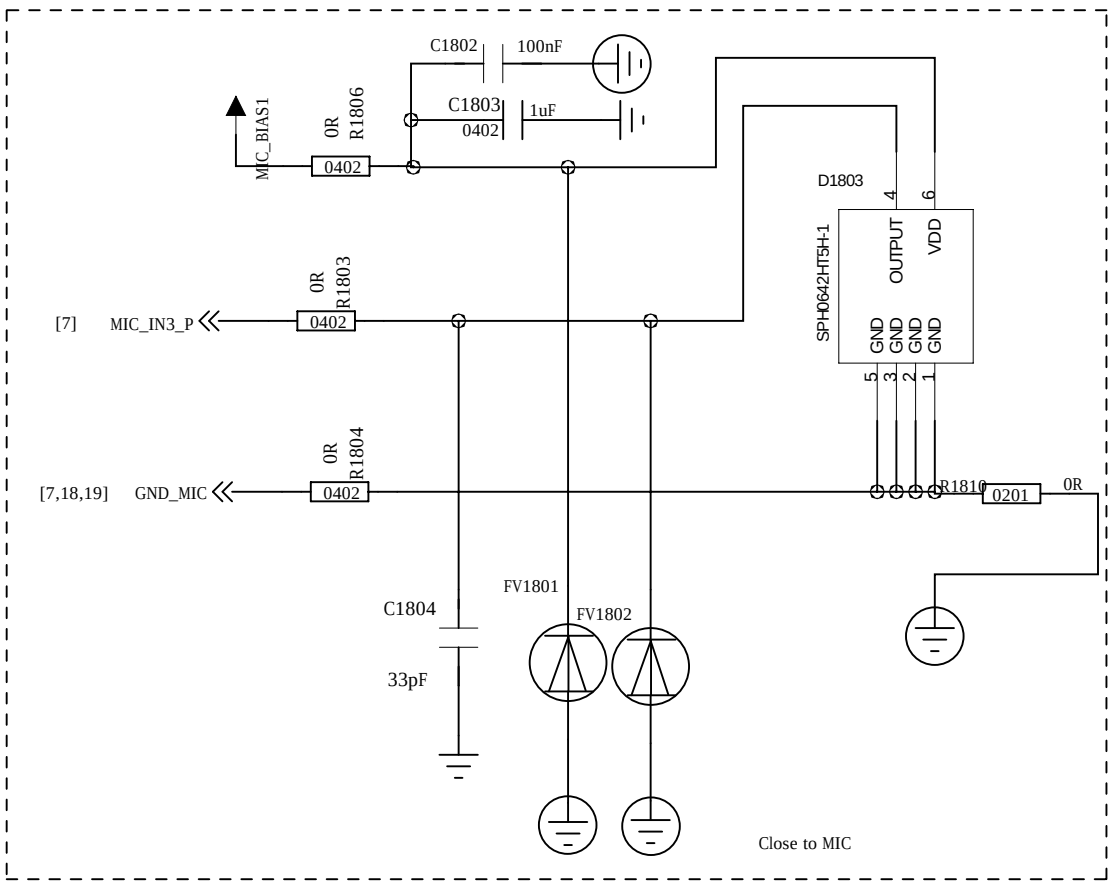
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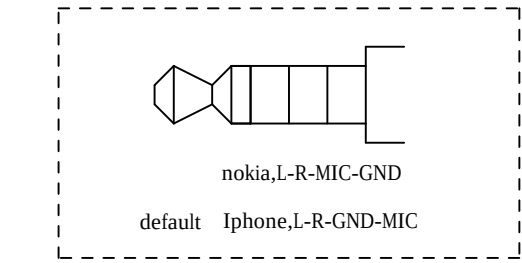
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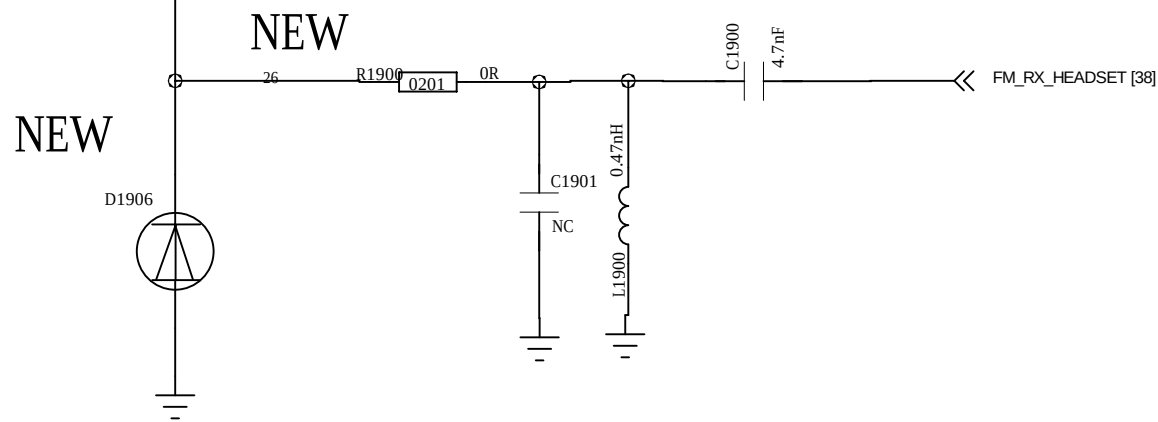
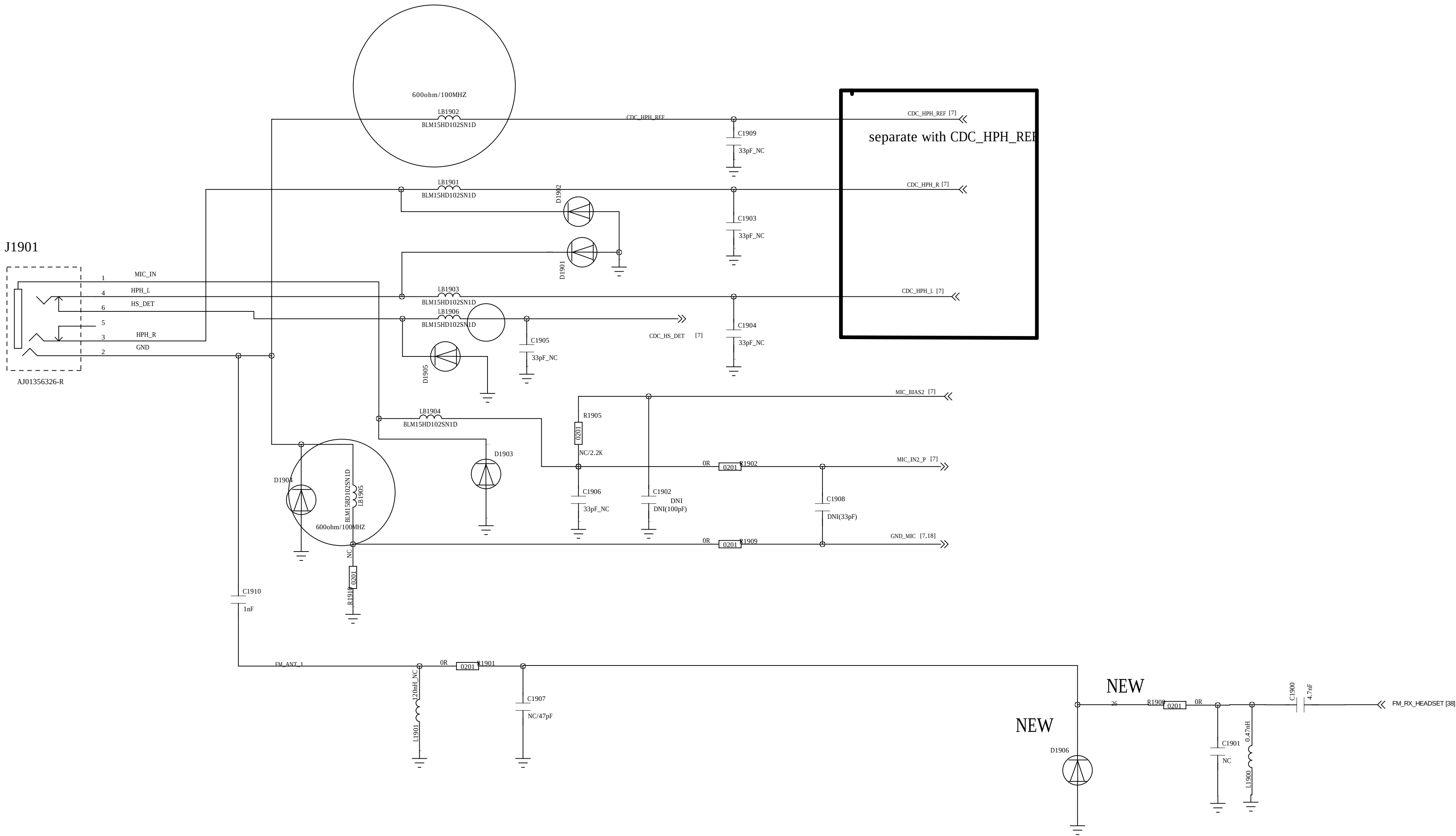
SECONDARY MIC



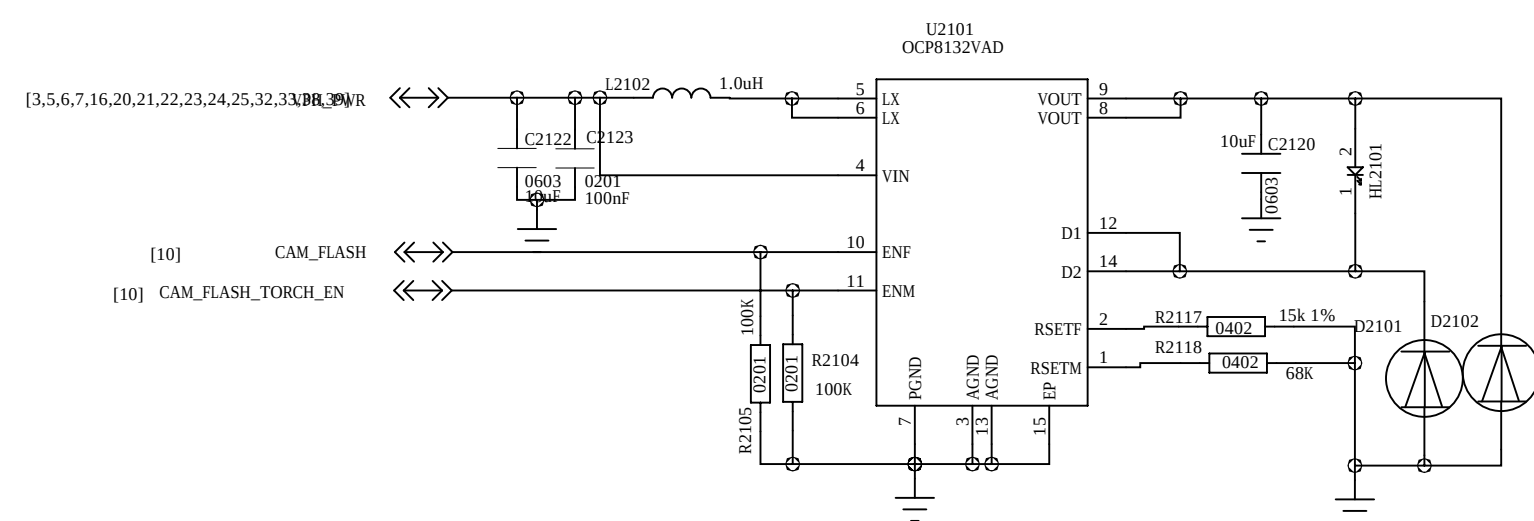
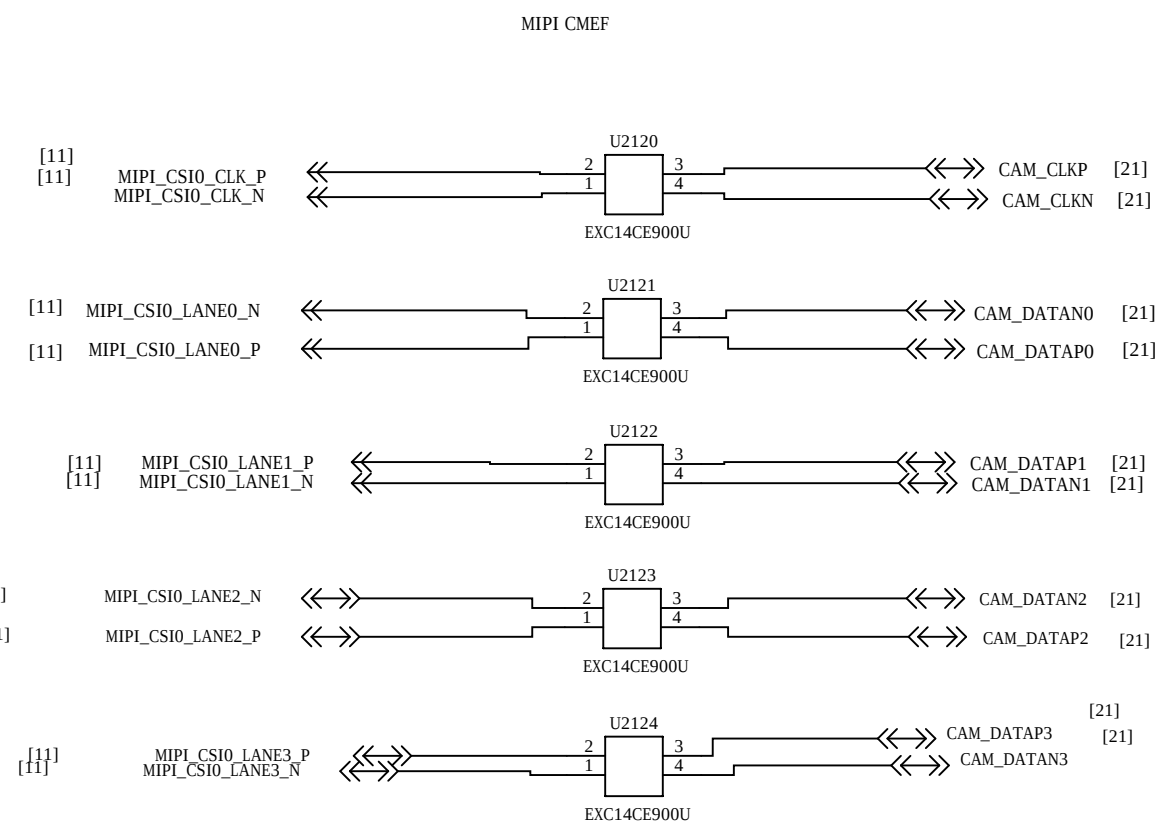
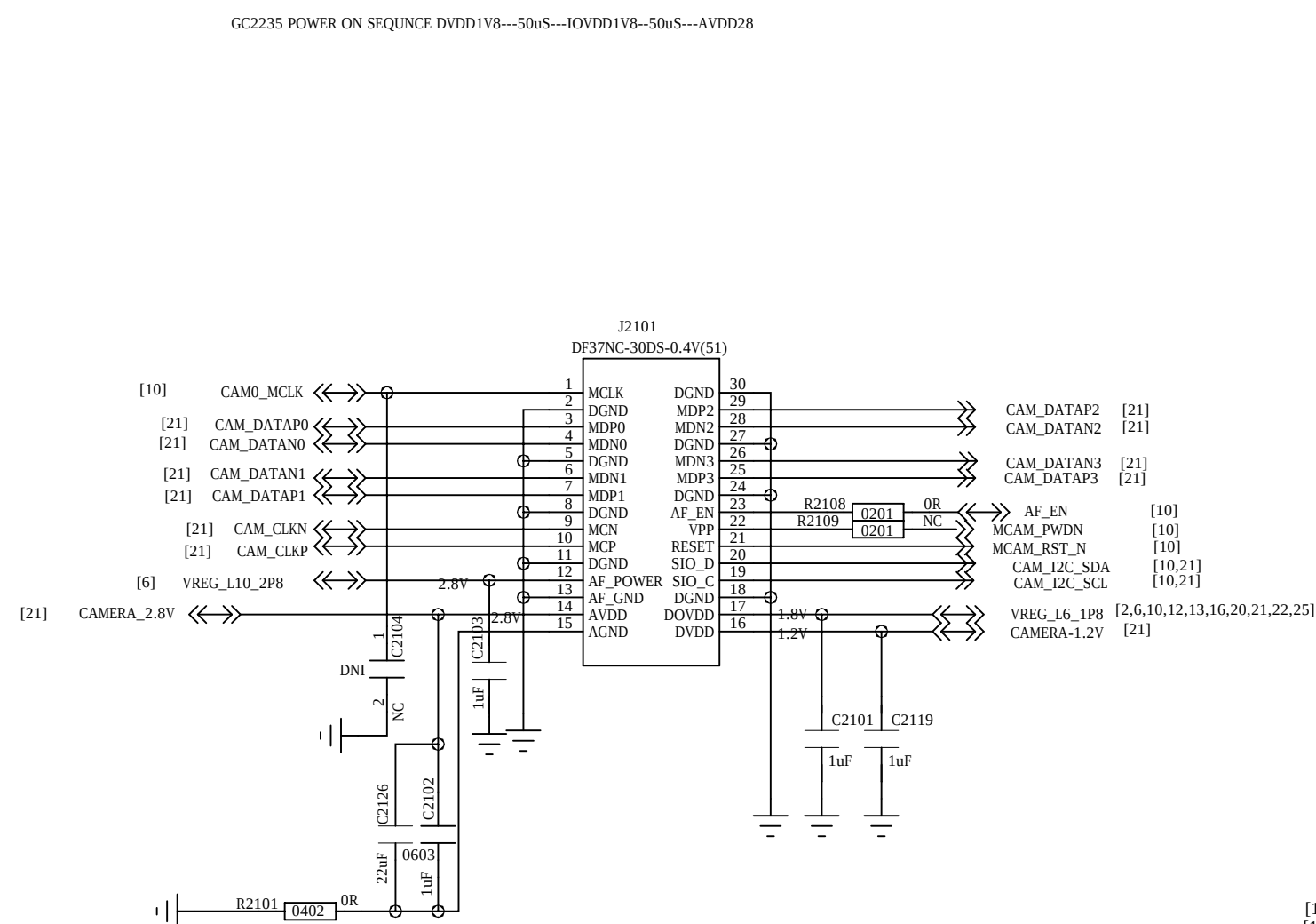
RECEIVER



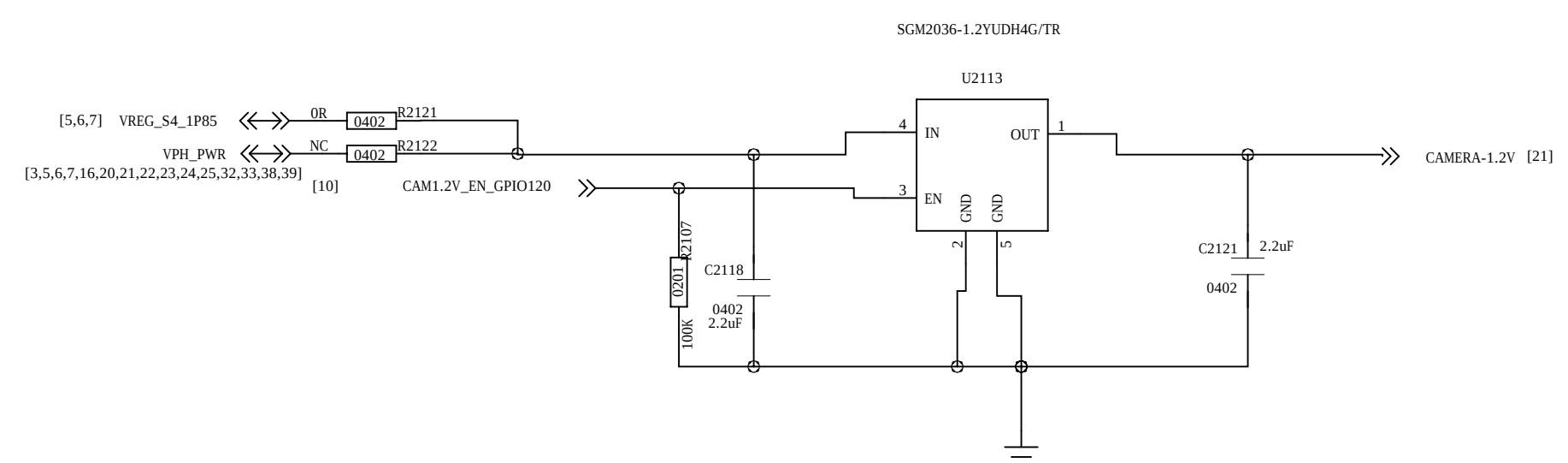
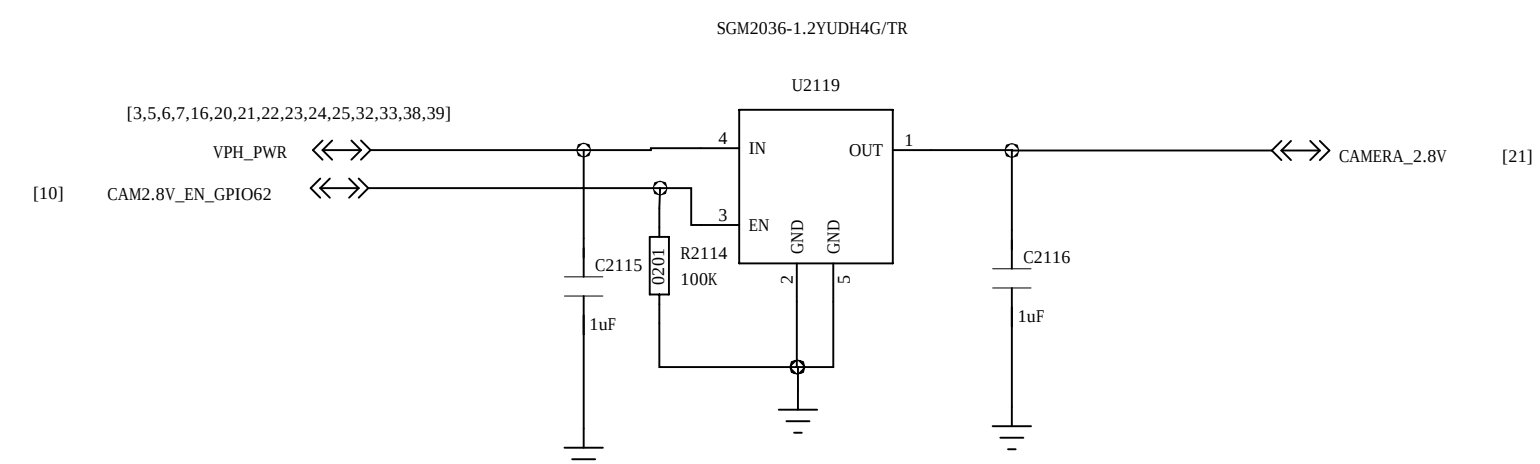
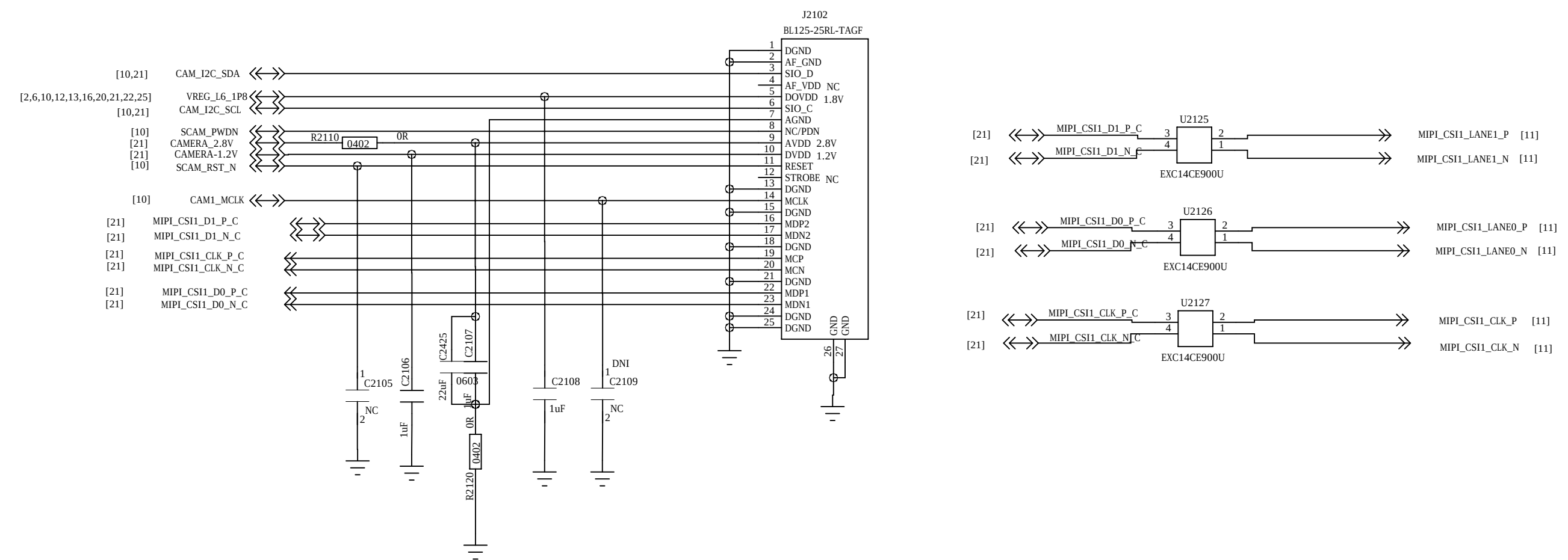
HEADSET



CS0:13M
CS1:5M



OCP8132 TORCH=Id1+Id2 Imovie= Id1+Id2=6800/R2118=200mA R2118=68K
FLASH=Id1+Id2 Iflash= Id1+Id2=6800/R2117=906mA R2117=15K



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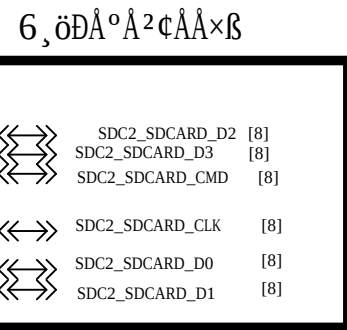
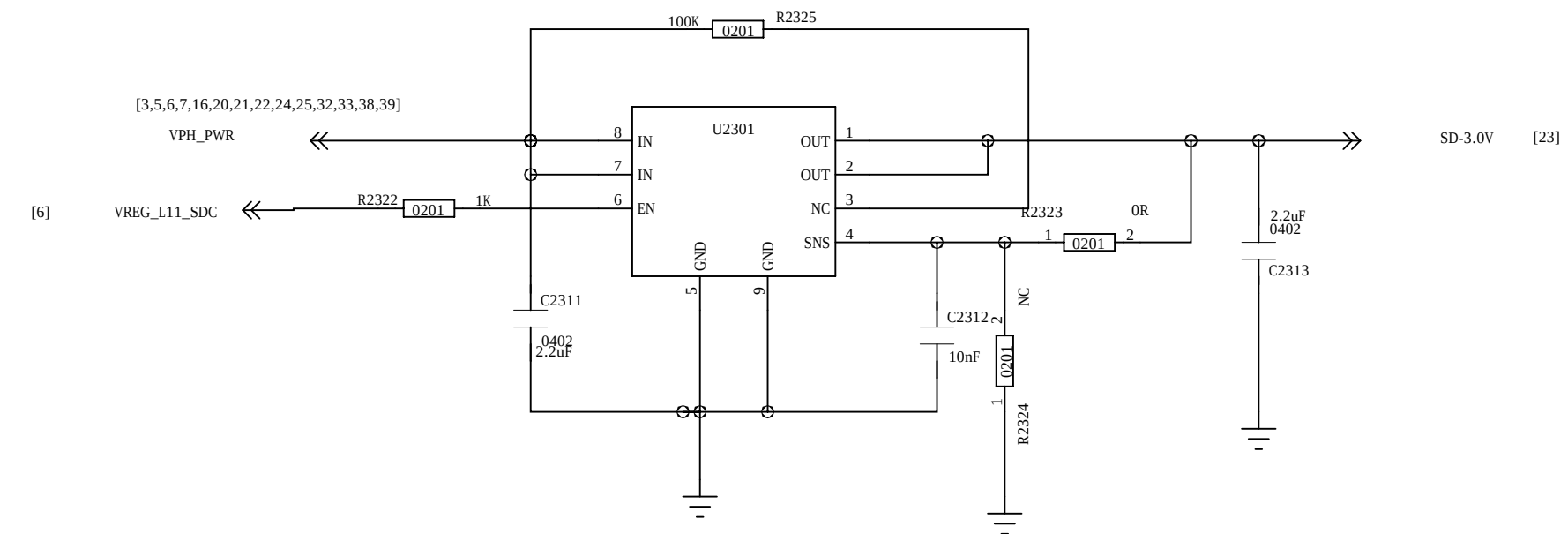
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FRM

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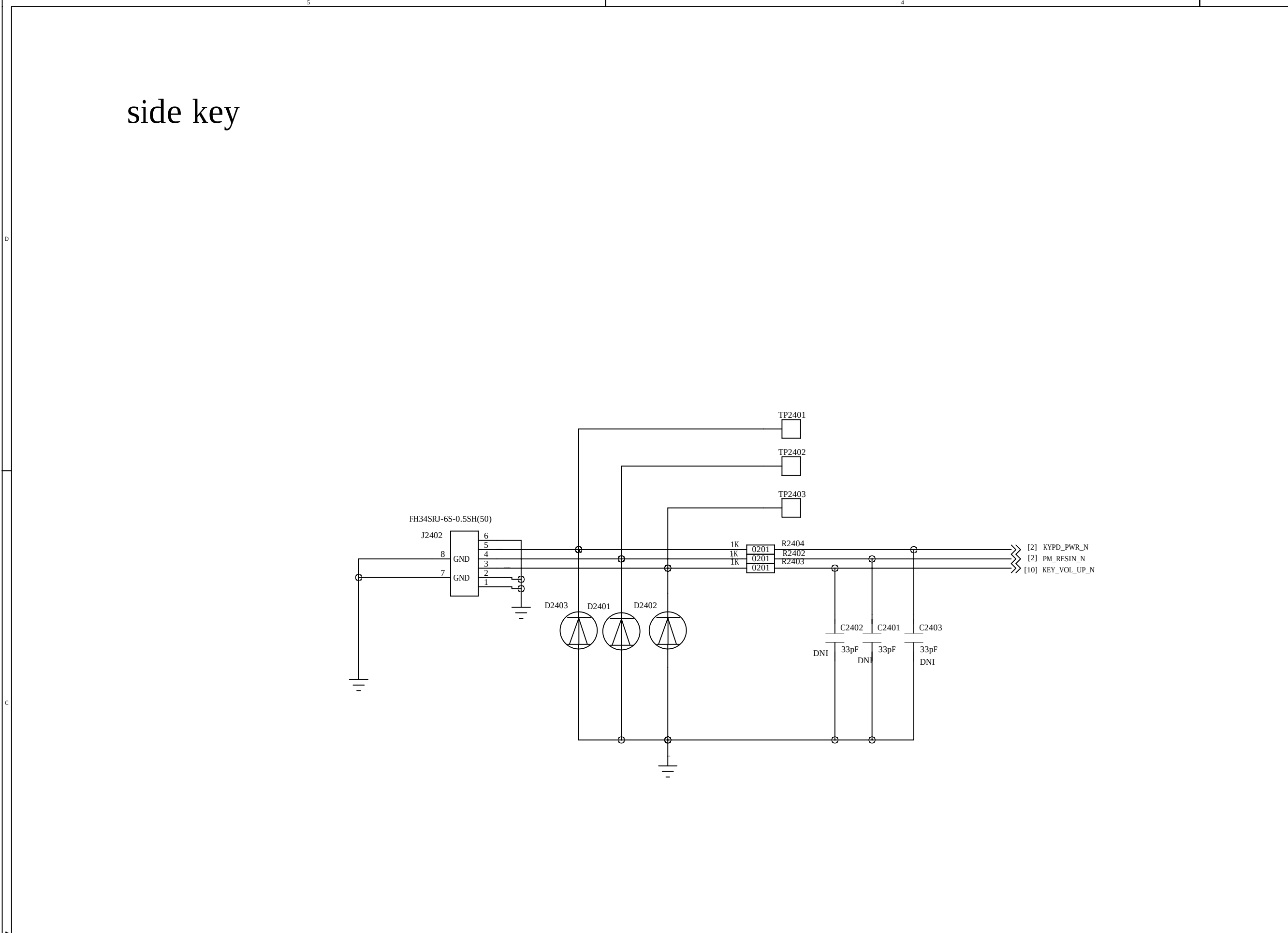
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side key



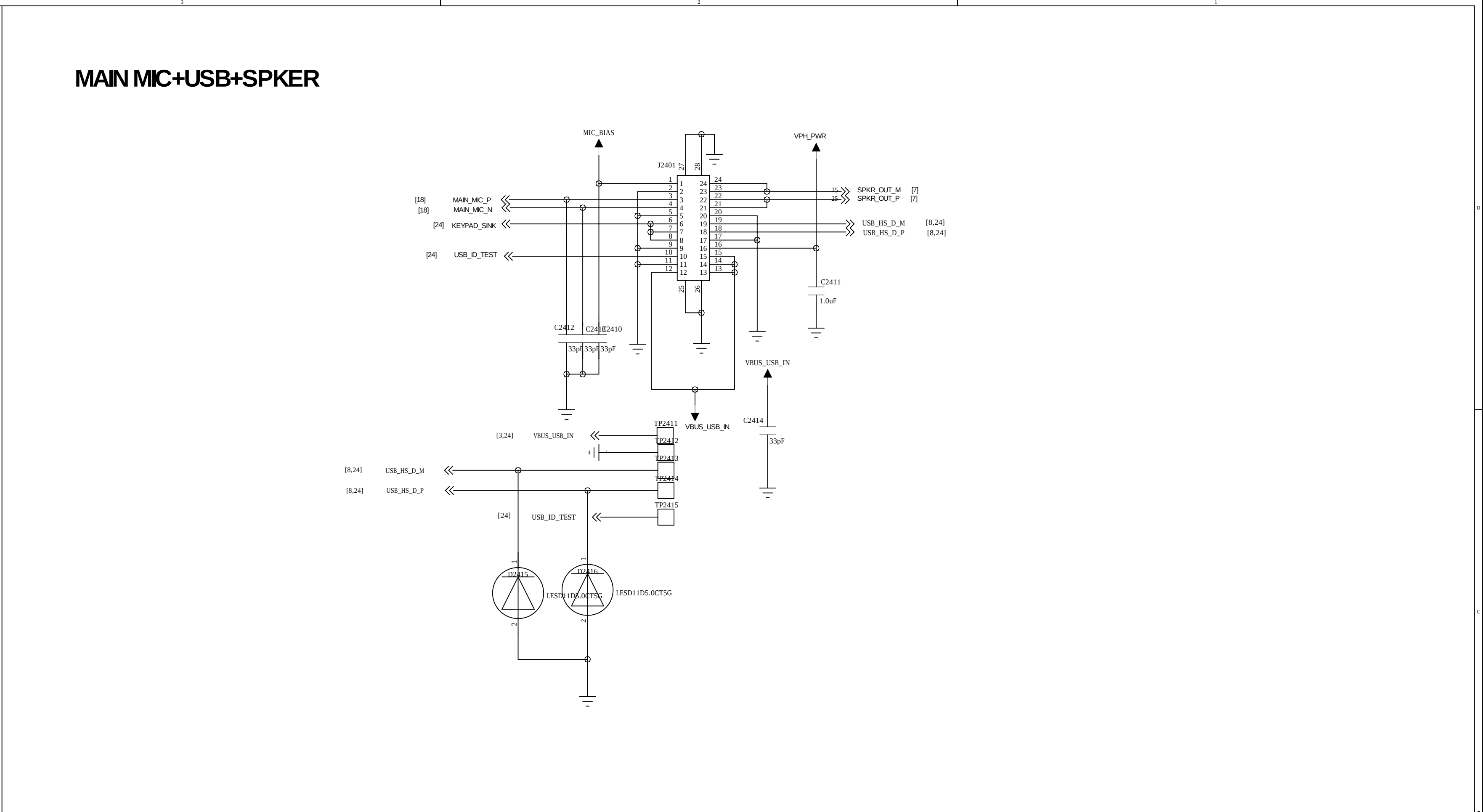
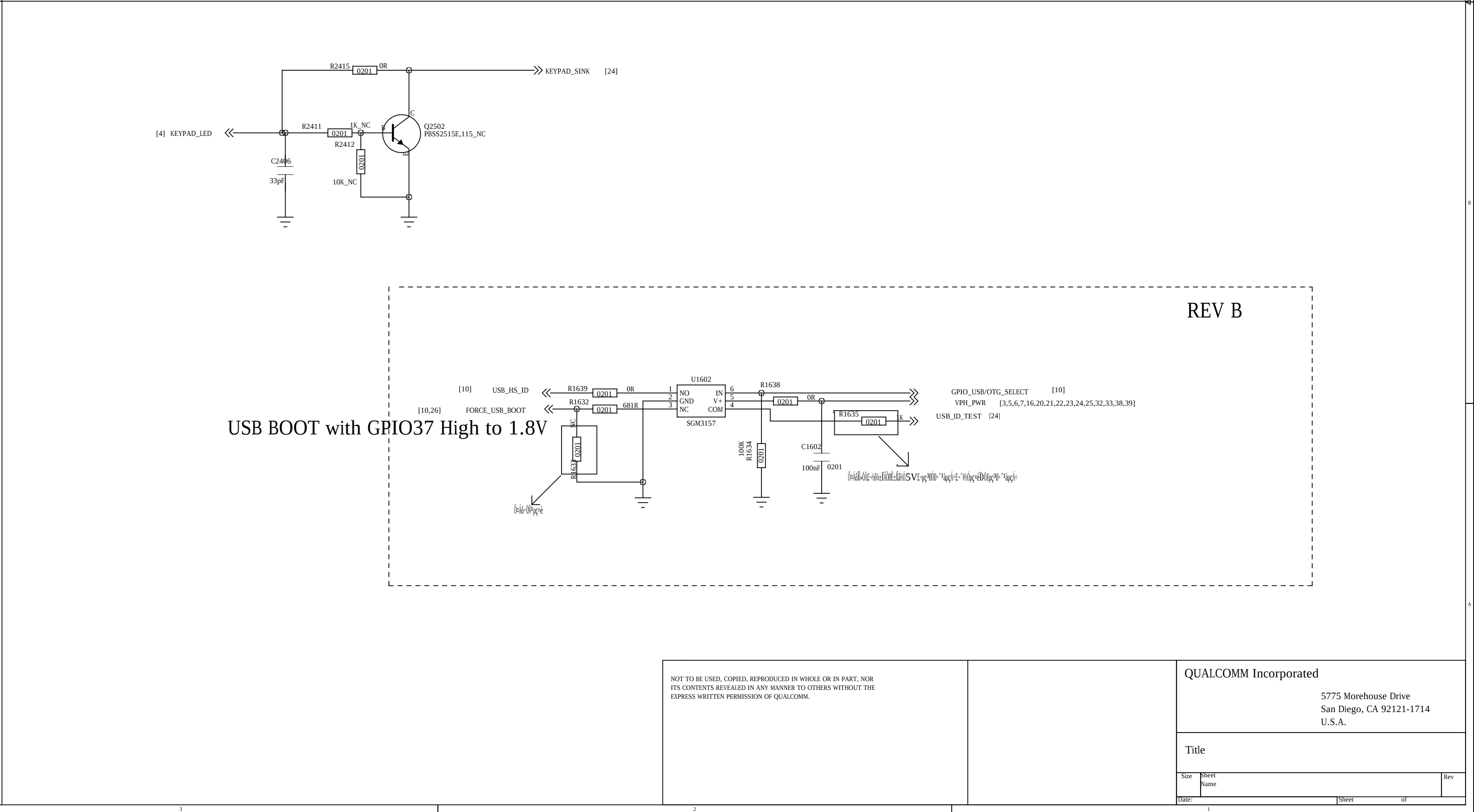
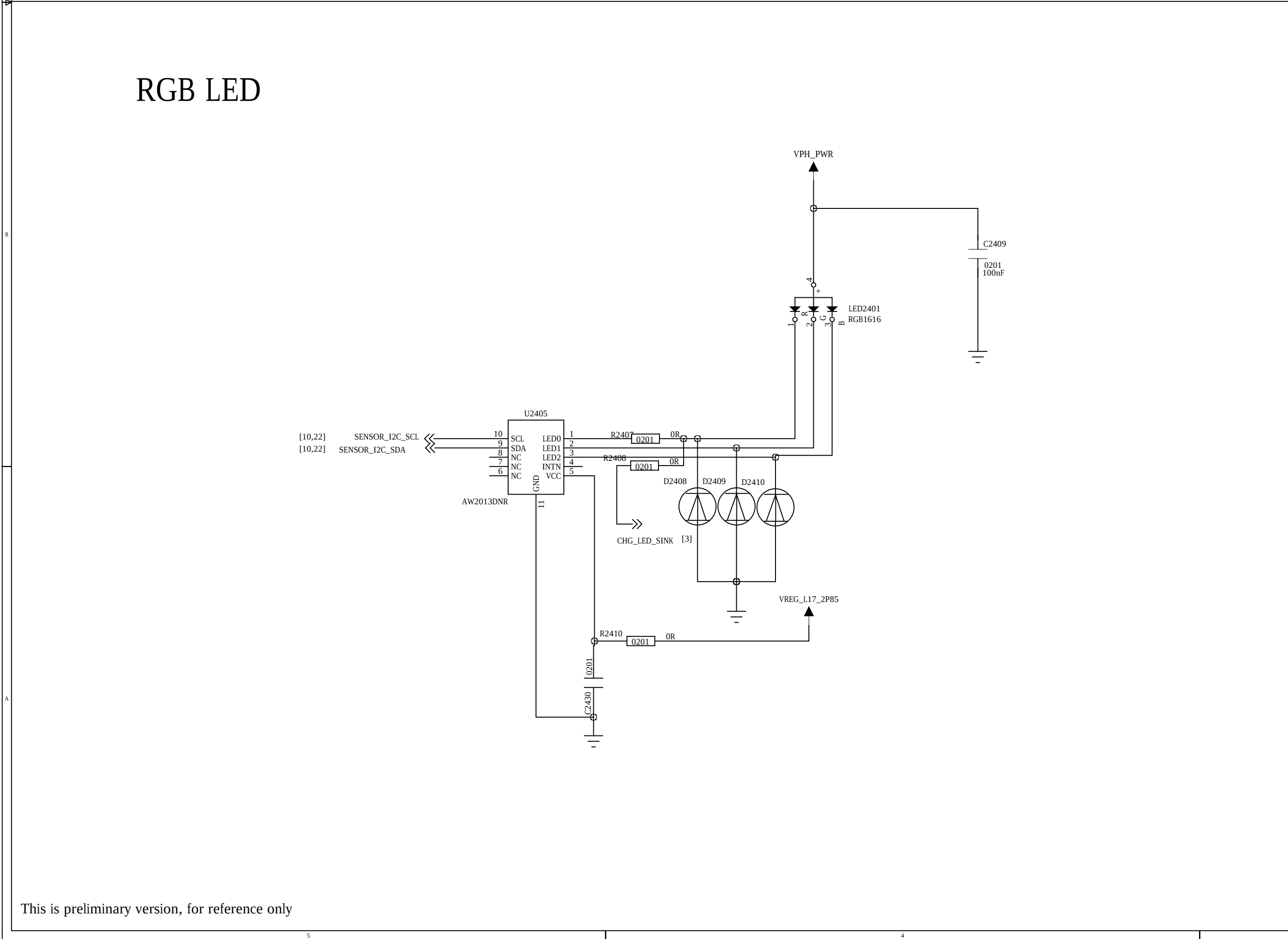
MAIN MIC+USB+SPKER

The schematic diagram illustrates the MAIN MIC+USB+SPKER interface circuit. The central component is a 24-pin connector, J2401. The connections are as follows:

- Pin 1:** MIC_BIAS
- Pin 2:** MAIN_MIC_P
- Pin 3:** MAIN_MIC_N
- Pin 4:** KEYPAD_SINK
- Pin 5:** USB_ID_TEST
- Pin 6:** USB_HS_D_M
- Pin 7:** USB_HS_D_P
- Pin 8:** USB_ID_TEST
- Pin 9:** VBUS_USB_IN
- Pin 10:** VBUS_USB_N
- Pin 11:** VBUS_USB_IN
- Pin 12:** VBUS_USB_N
- Pin 13:** SPKR_OUT_M
- Pin 14:** SPKR_OUT_P
- Pin 15:** VPH_PWR
- Pin 16:** VPH_PWR
- Pin 17:** SPKR_OUT_M
- Pin 18:** SPKR_OUT_P
- Pin 19:** VPH_PWR
- Pin 20:** VPH_PWR
- Pin 21:** VPH_PWR
- Pin 22:** VPH_PWR
- Pin 23:** VPH_PWR
- Pin 24:** VPH_PWR

Key components and their values:

- Capacitors:** C2412 (33pF), C2410 (33pF), C2411 (1.0uF), C2414 (33pF).
- Transformer:** TP2411.
- Diodes:** 1ESD11D5.0CTSG.
- Resistor:** 10k.

[illegible]

Schematic Diagram 1: Keypad LED and Sink

This schematic shows the connection for a keypad LED and its sink. The LED is connected to a keypad pad through resistor R2411 (0.020). A 33pF capacitor C2446 is connected to ground. The other end of the LED is connected to a keypad pad through resistor R2412 (0.020) and a 10K_NC resistor. The sink is connected to a keypad pad through resistor R2415 (0.020).

Schematic Diagram 2: USB Boot with GPIO37 High to 1.8V

This schematic shows the connection for a USB boot circuit. It includes resistors R1639 (0.020), R1632 (0.020), R1638 (0.020), and R1635 (0.020). A 100K resistor is connected to ground. A 100nF capacitor C1602 is connected to ground. The circuit is powered by VPH_PWR [3,5,6,7,16,20,21,22,23,24,25,32,33,38,39]. The output is connected to a keypad pad through resistor R1635 (0.020).

Table 1: Component Values

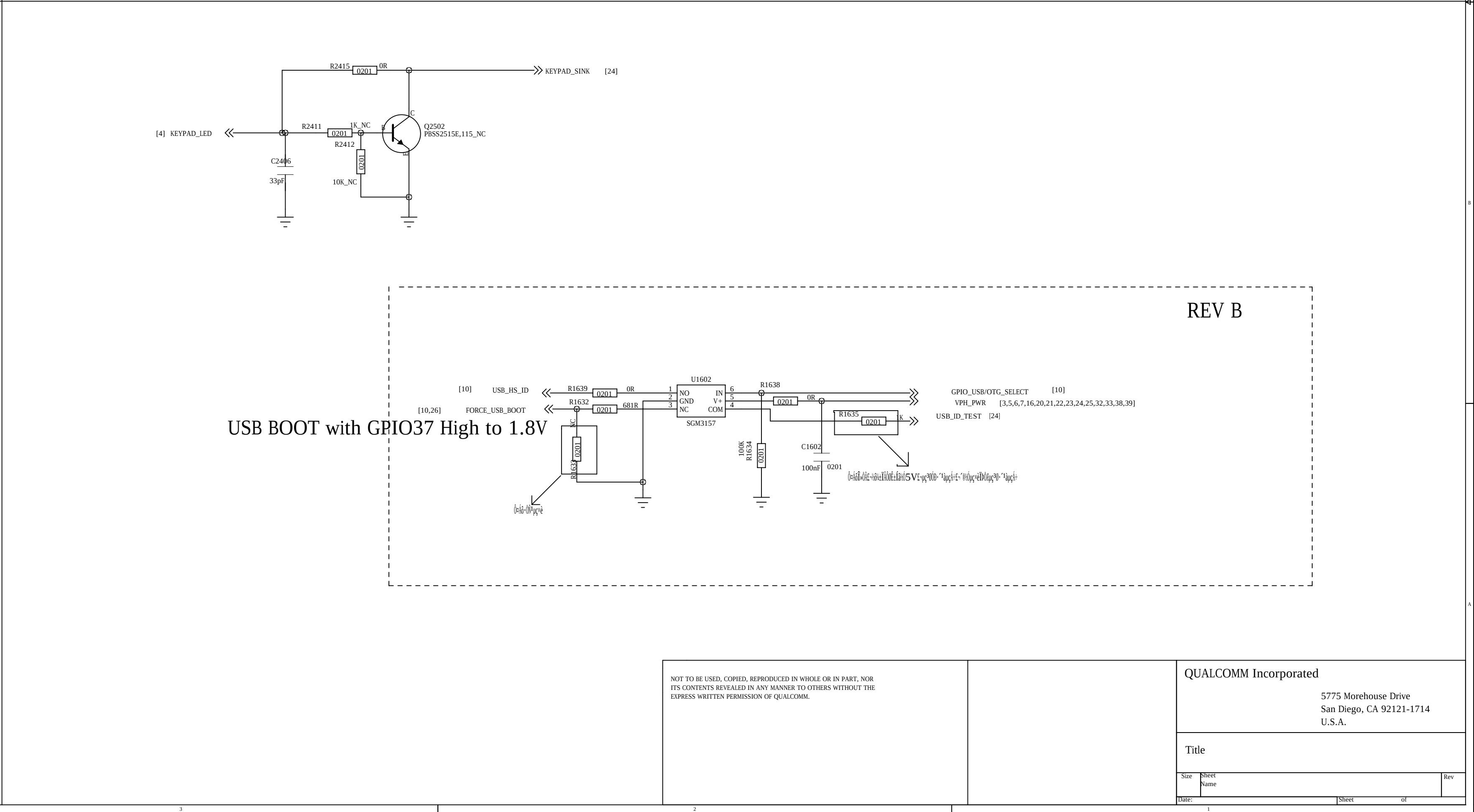
Component	Value
R2411	0.020
C2446	33pF
R2412	0.020
10K_NC	10K
R2415	0.020
R1639	0.020
R1632	0.020
R1638	0.020
R1635	0.020
100K	100K
C1602	100nF

Table 2: Signal Connections

Signal	Pin	Value
KEYPAD_LED	[4]	KEYPAD_LED
KEYPAD_SINK	[24]	KEYPAD_SINK
GPIO_USB_OTG_SELECT	[10]	GPIO_USB_OTG_SELECT
VPH_PWR	[3,5,6,7,16,20,21,22,23,24,25,32,33,38,39]	VPH_PWR
USB_ID_TEST	[24]	USB_ID_TEST

Table 3: Revision History

Rev	Date	Sheet	of	Rev
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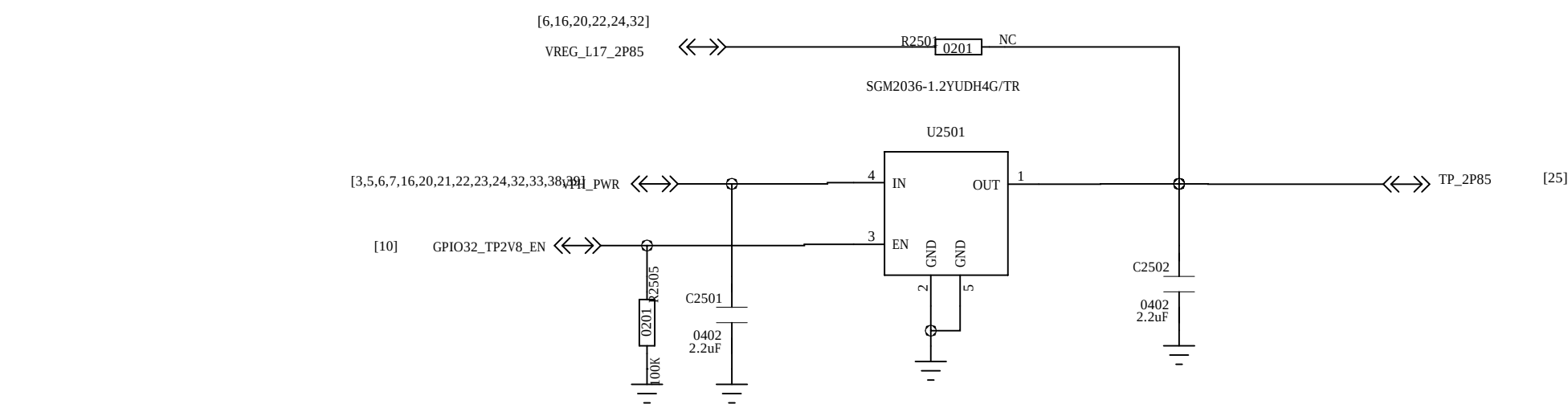
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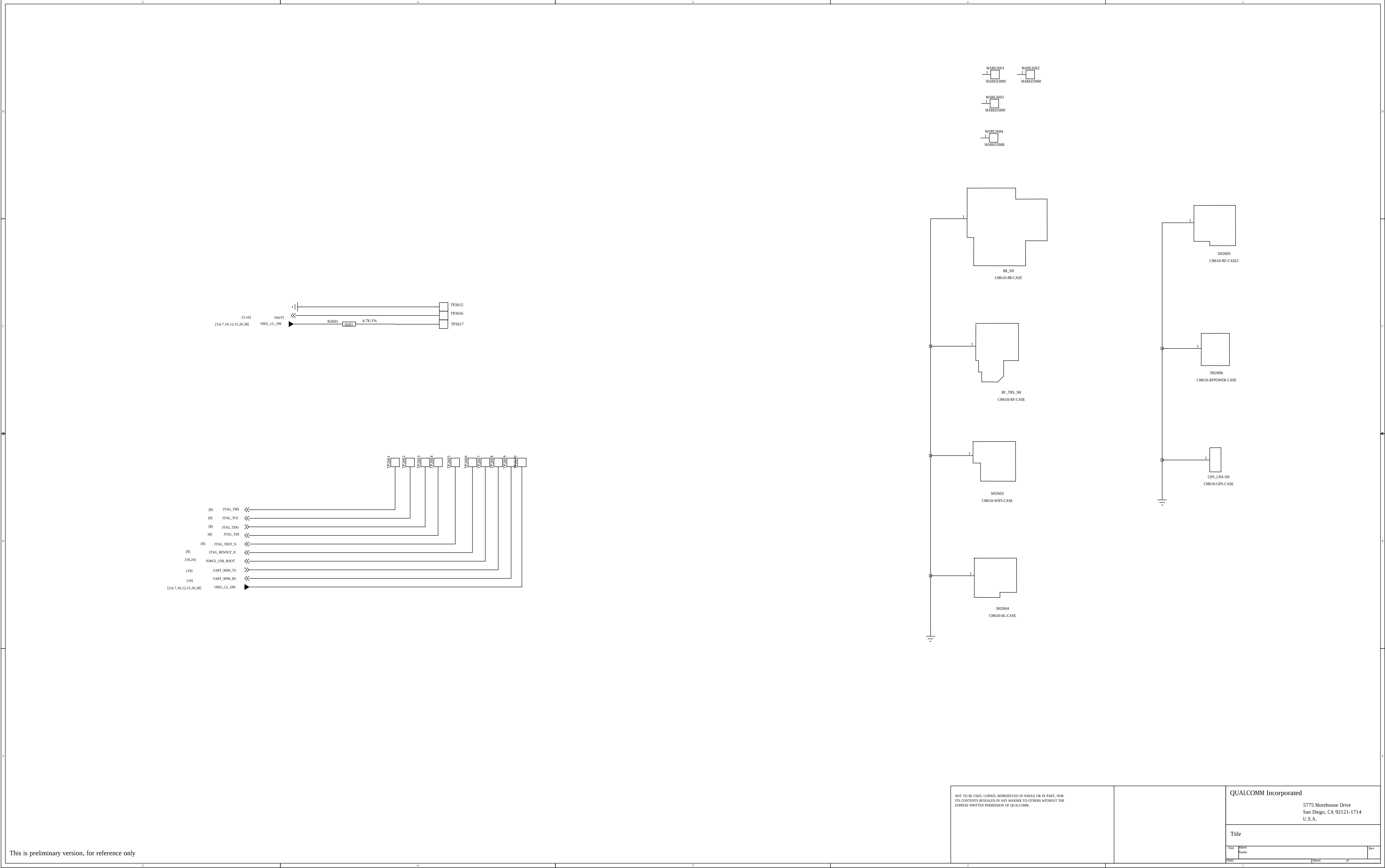


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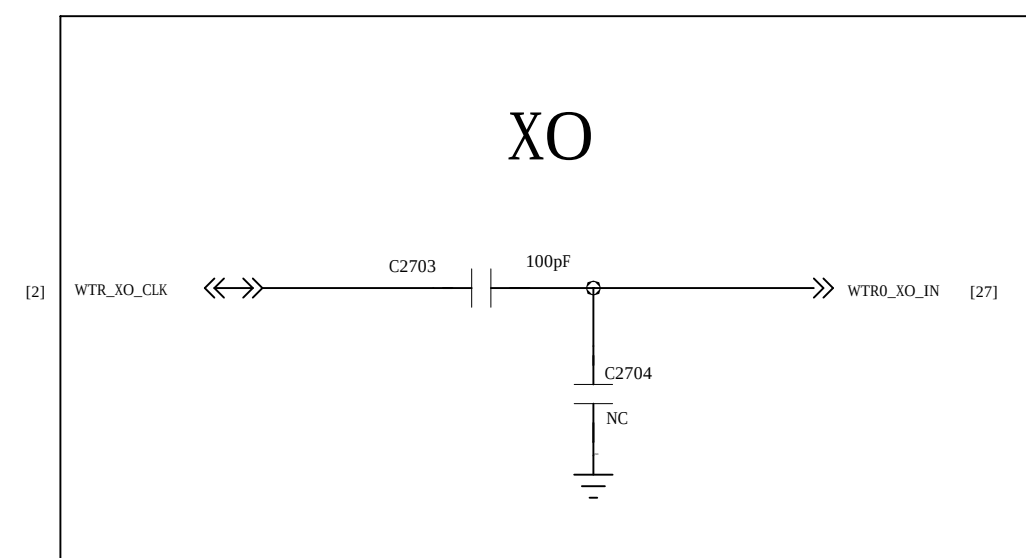
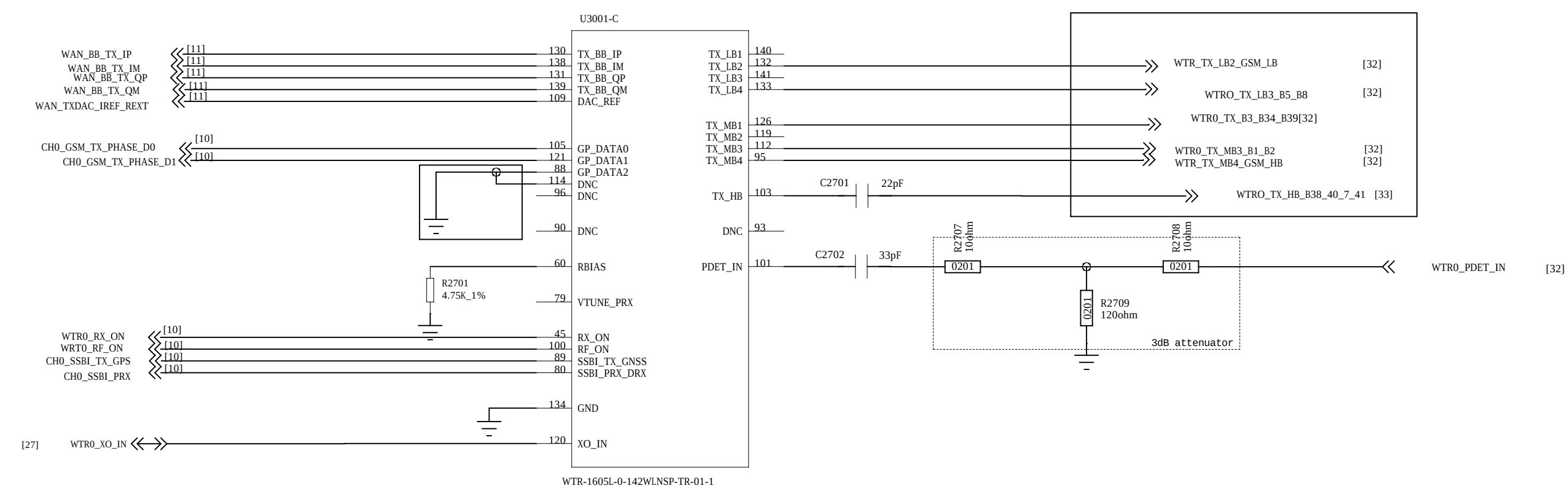
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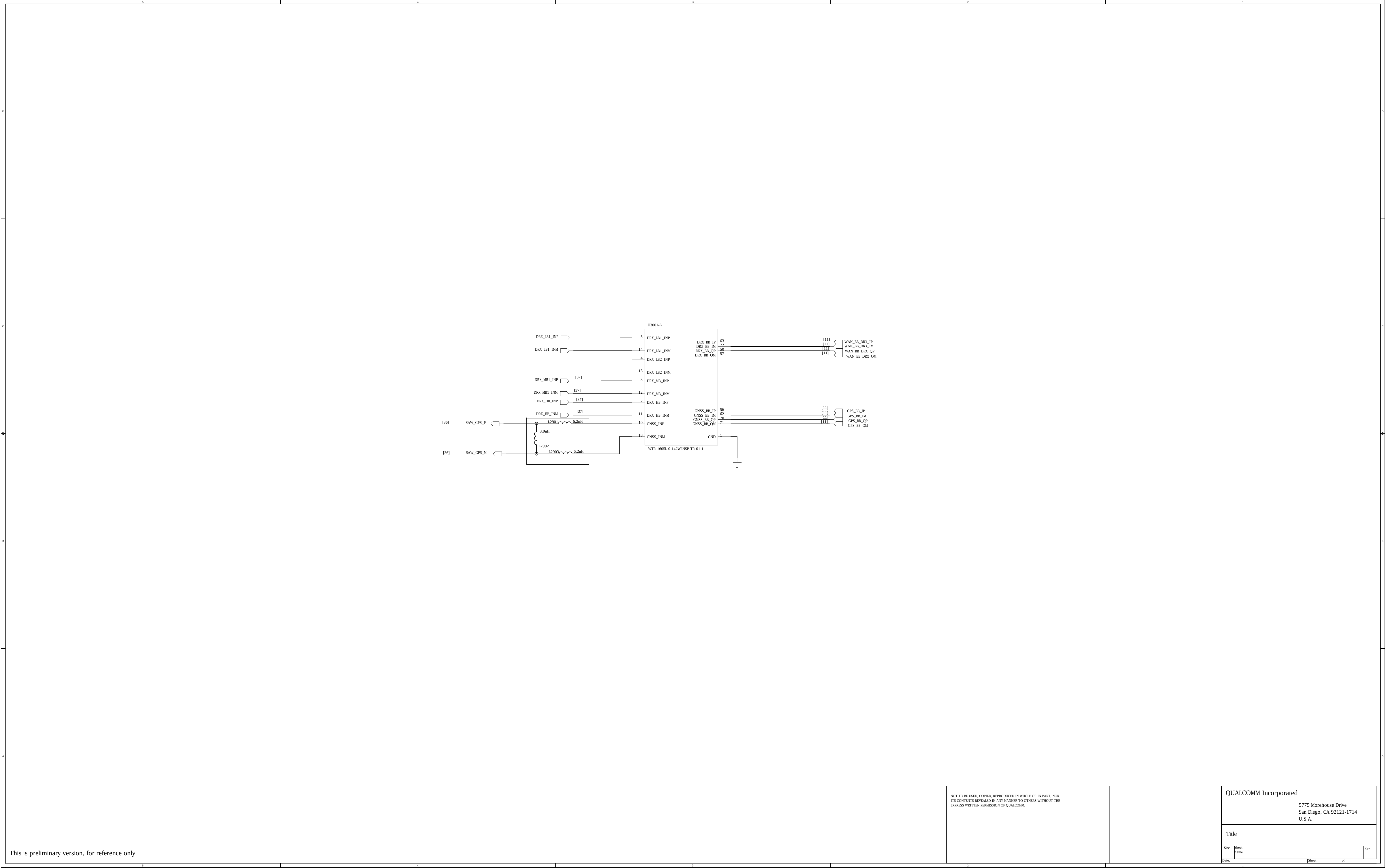
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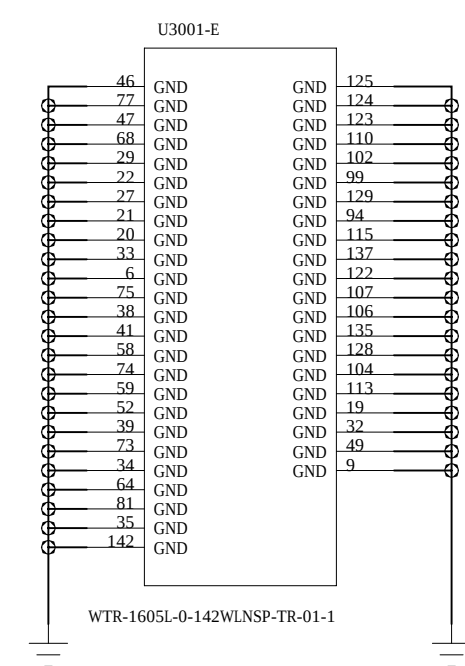
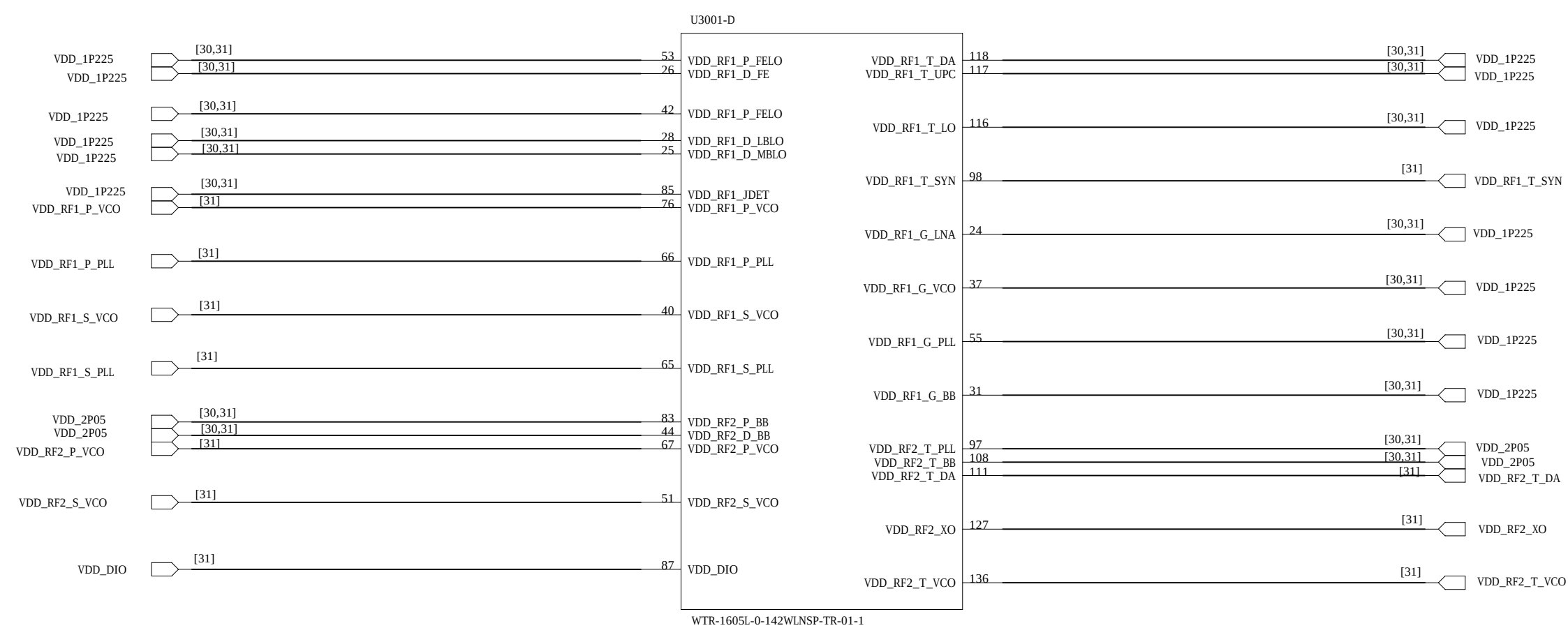
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2	1	



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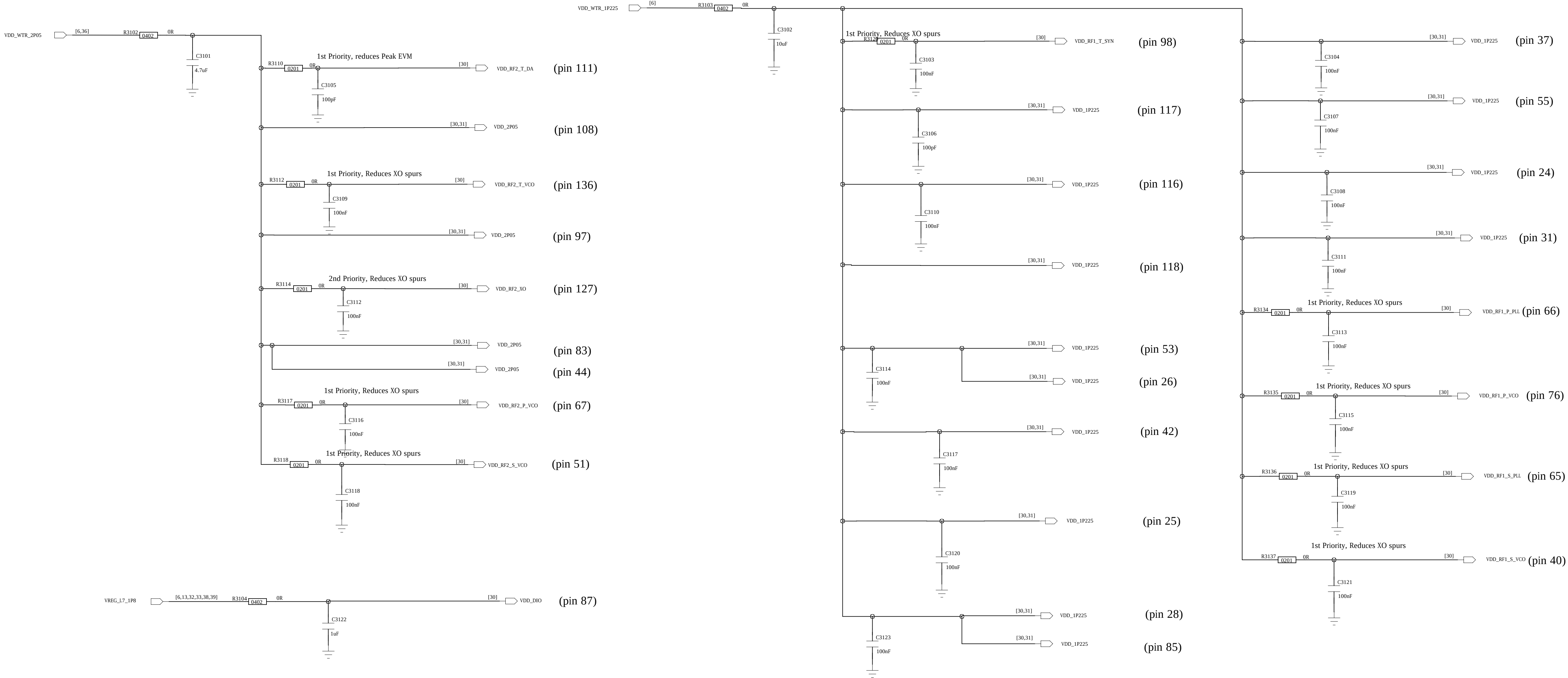
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Date:	Sheet of	

RF1 = 1.225V
RF2 = 2.05V
VDD_DIO = 1.8V



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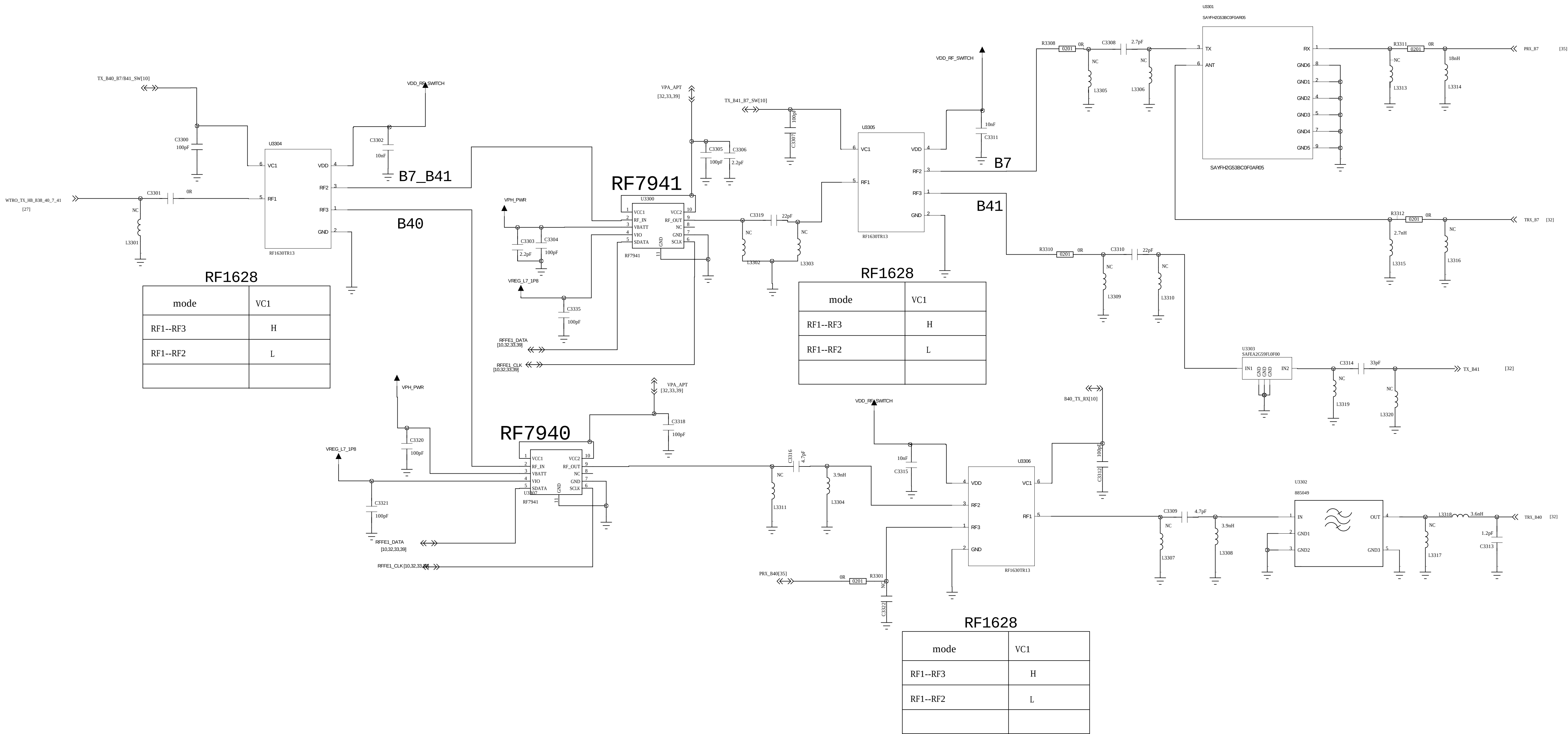
CTRL	STATE
0	RFC---RF1(pin1)
1	RFC---RF2(pin9)

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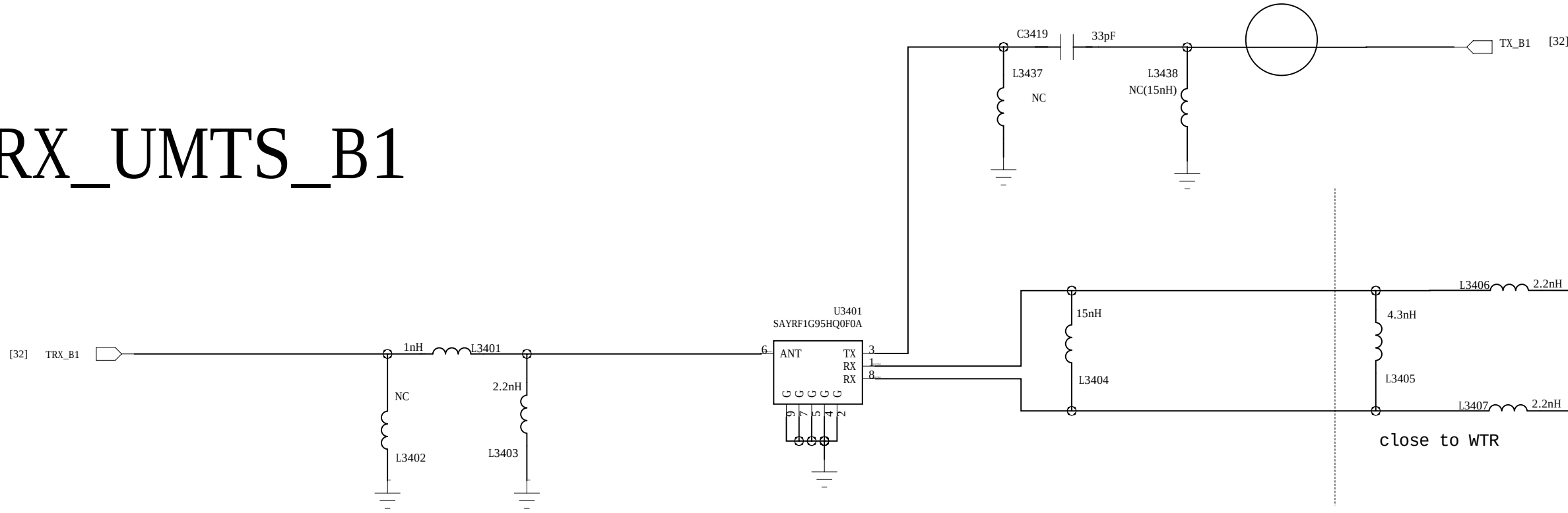
Title

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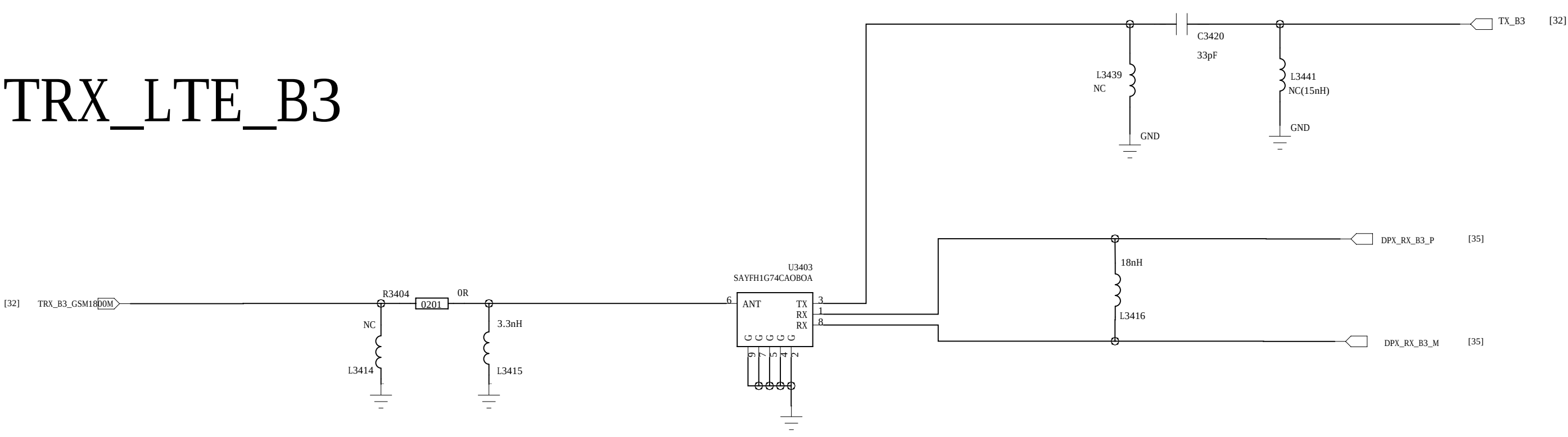
RF7940/RF7941_B7/40/41



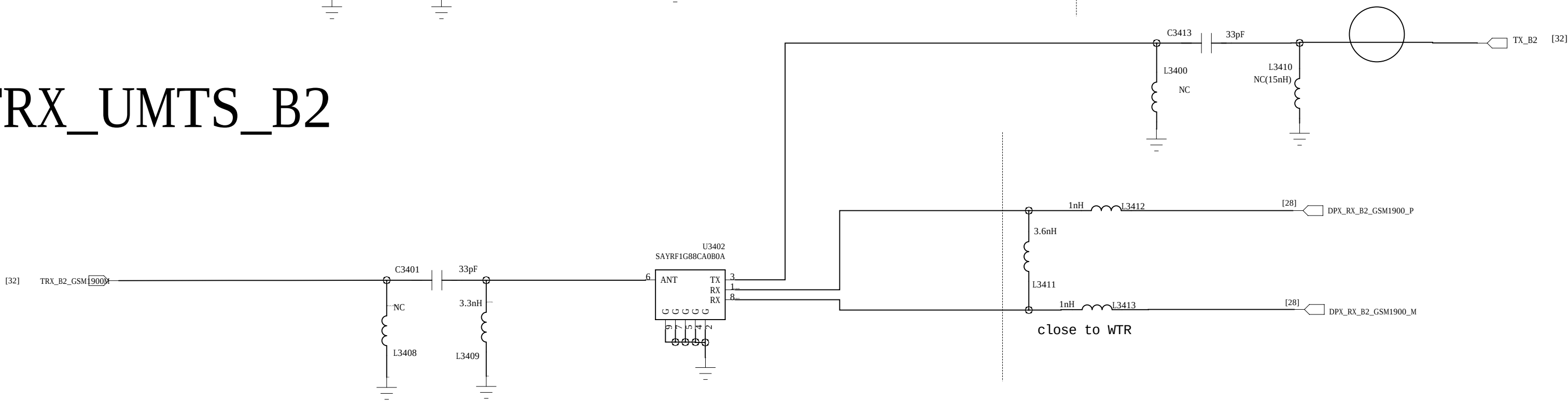
RX_UMTS_B1



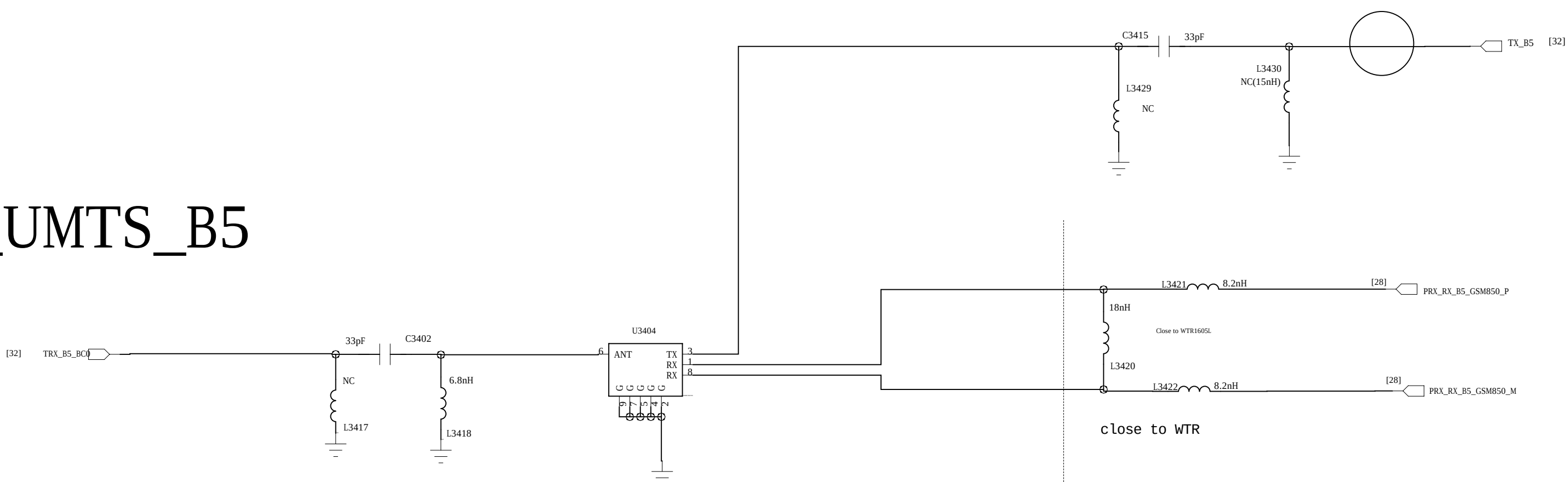
TRX_LTE_B3



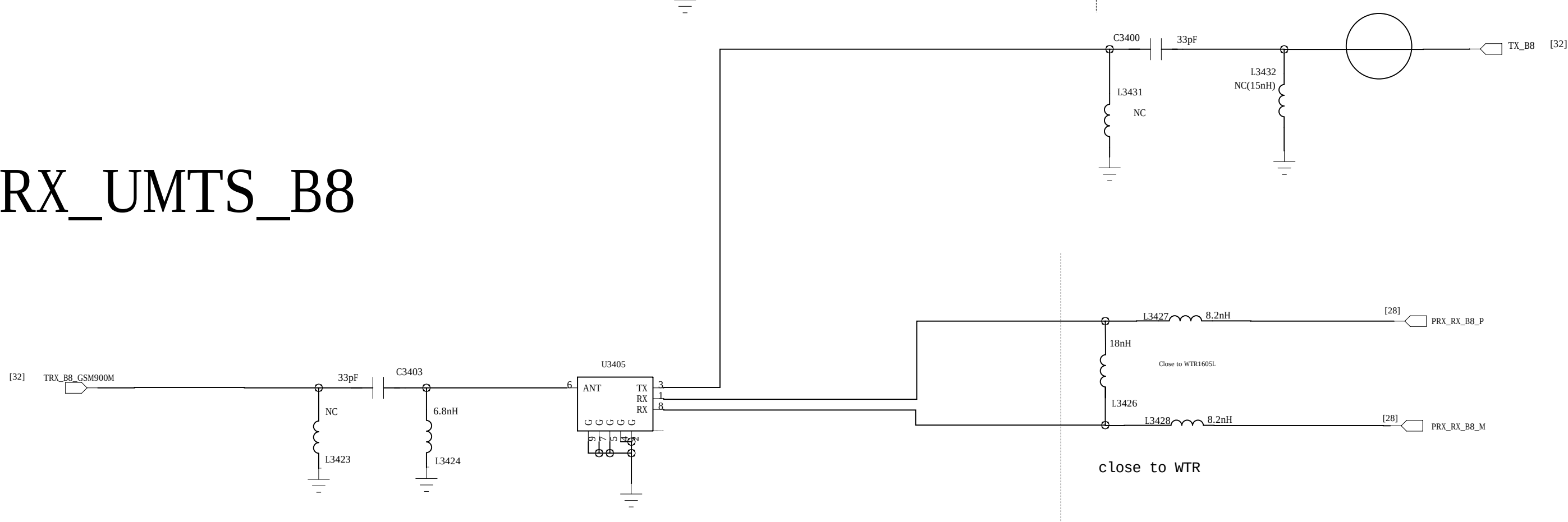
TRX_UMTS_B2



TRX_UMTS_B5

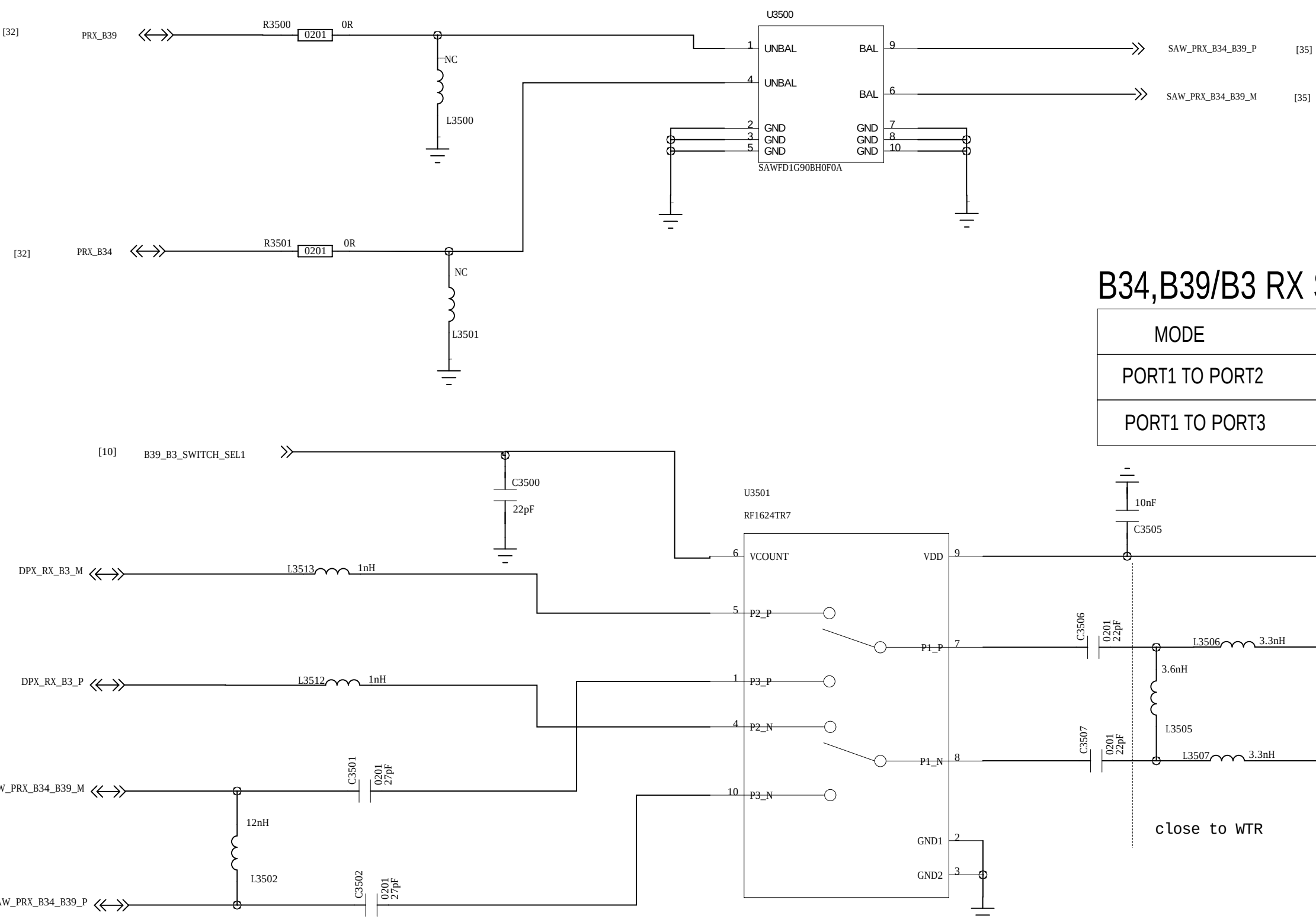


TRX_UMTS_B8



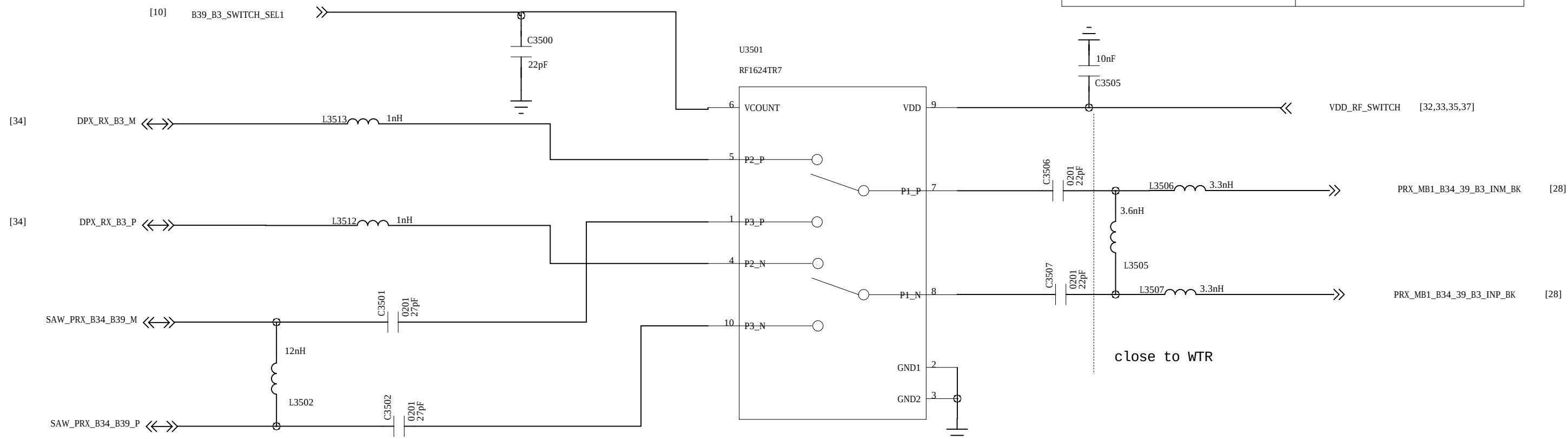
PRX_B34/B39

4/10 modif



B34,B39/B3 RX Switch

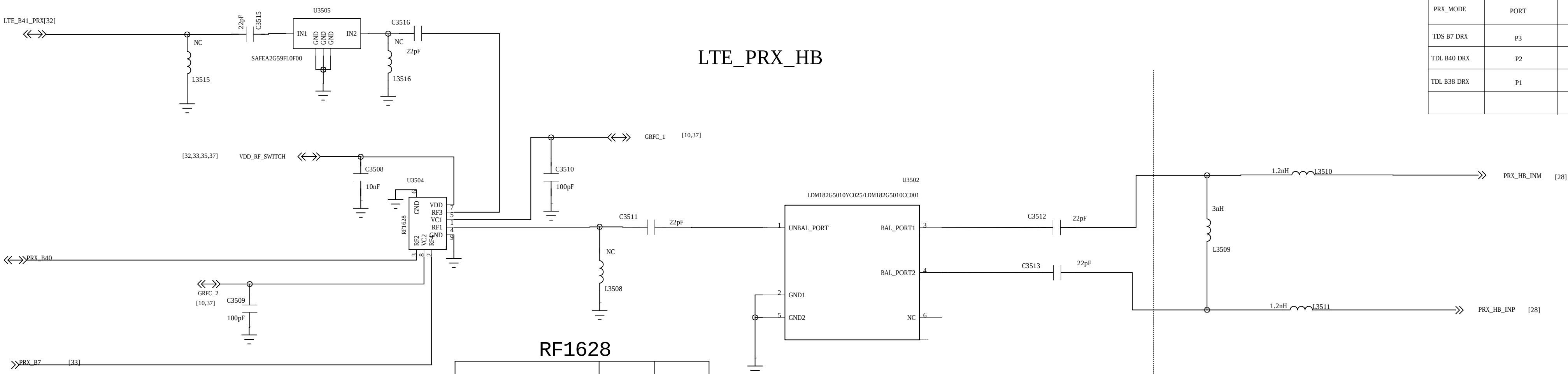
MODE	B39_B3_SWITCH_SEL
PORT1 TO PORT2	L
PORT1 TO PORT3	H



PRX_B7,B38,B40,Switch Logic

PRX_MODE	PORT	GRFC_1	GRFC_2
TDS B7 DRX	P3	H	L
TDL B40 DRX	P2	L	H
TDL B38 DRX	P1	L	L

LTE_PRX_HB



RF1628		
mode	VCTL1	VCTL2
RF1--RF3	L	H
RF1--RF2	H	L
RF1--RF4	H	H

close to WTR

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BYPASS LNA&FILTER



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Size	Sheet Name	Rev
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Date:	Sheet	of
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RF1628		
mode	VCTL1	VCTL2
RF1--RF3	L	H
RF1--RF2	H	L
RF1--RF4	H	H

LTE B1_B39_B3 RX Switch

RF1628	
mode	VC1
RF1--RF3	H
RF1--RF2	L

RF1628

mode	VCTL1	VCTL2
RF1--RF3	L	H
RF1--RF2	H	L
RF1--RF4	H	H

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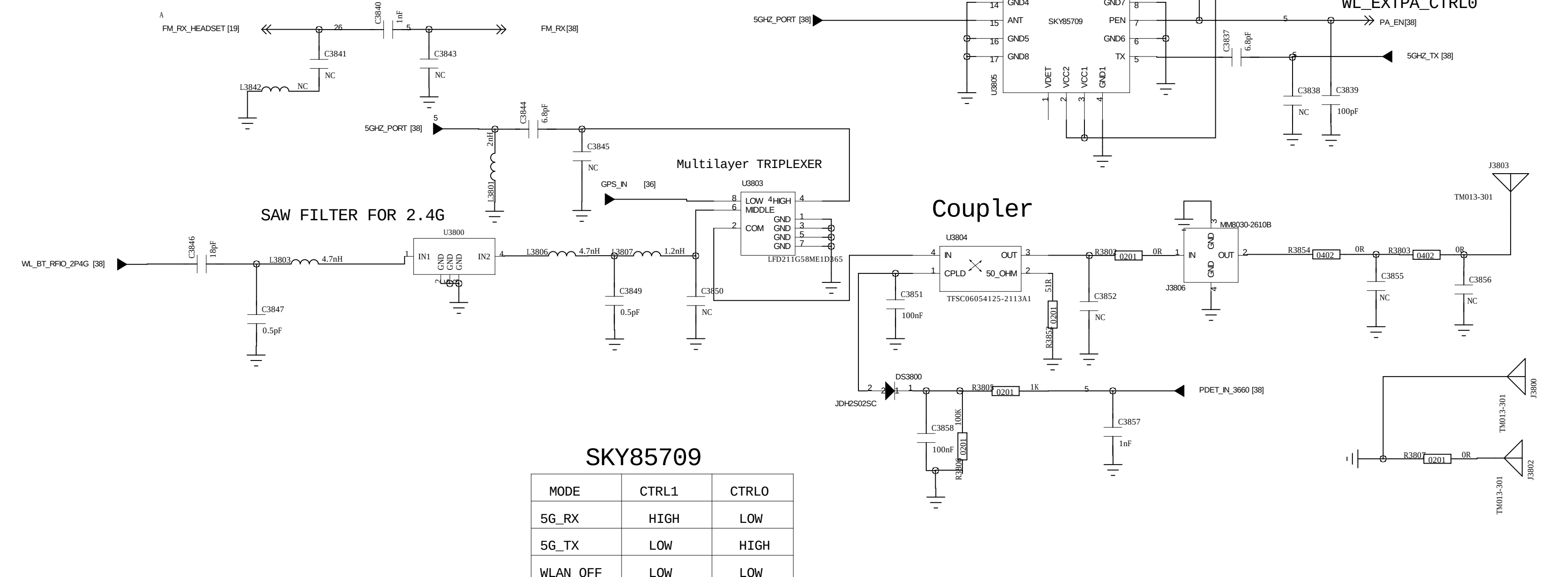
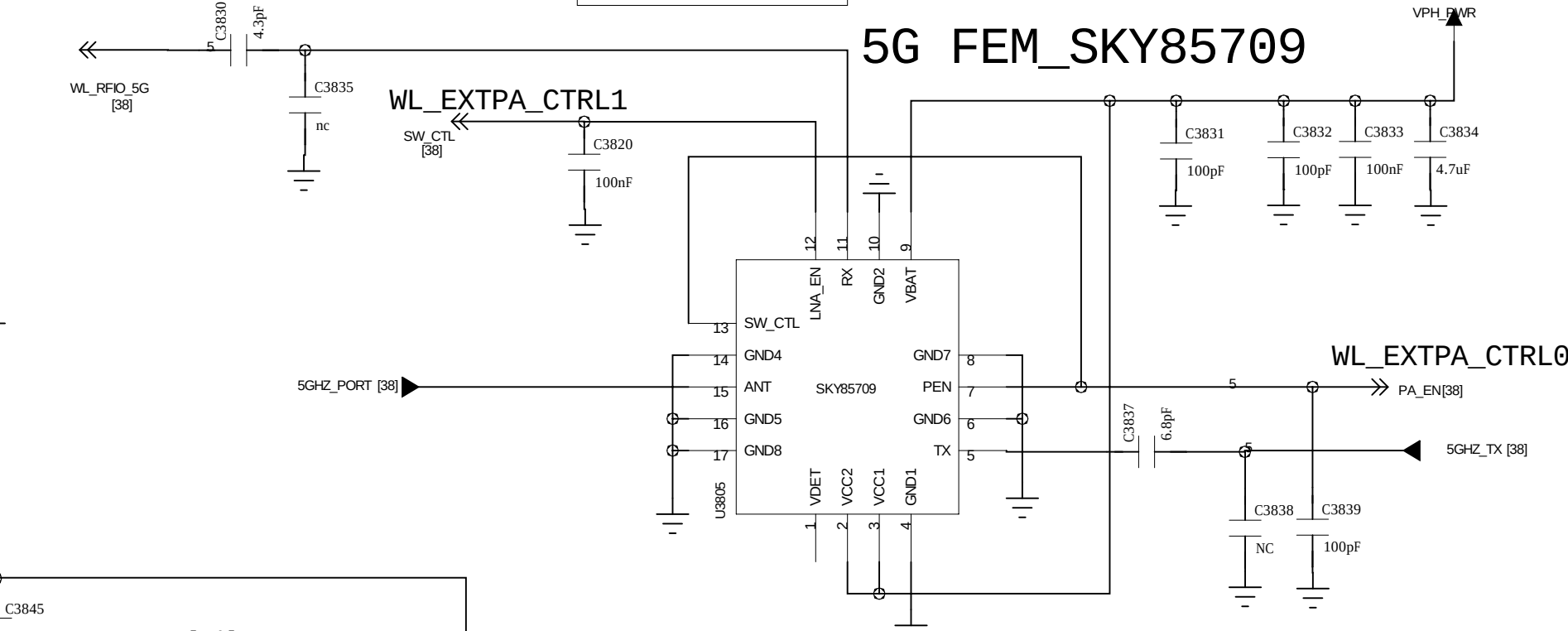
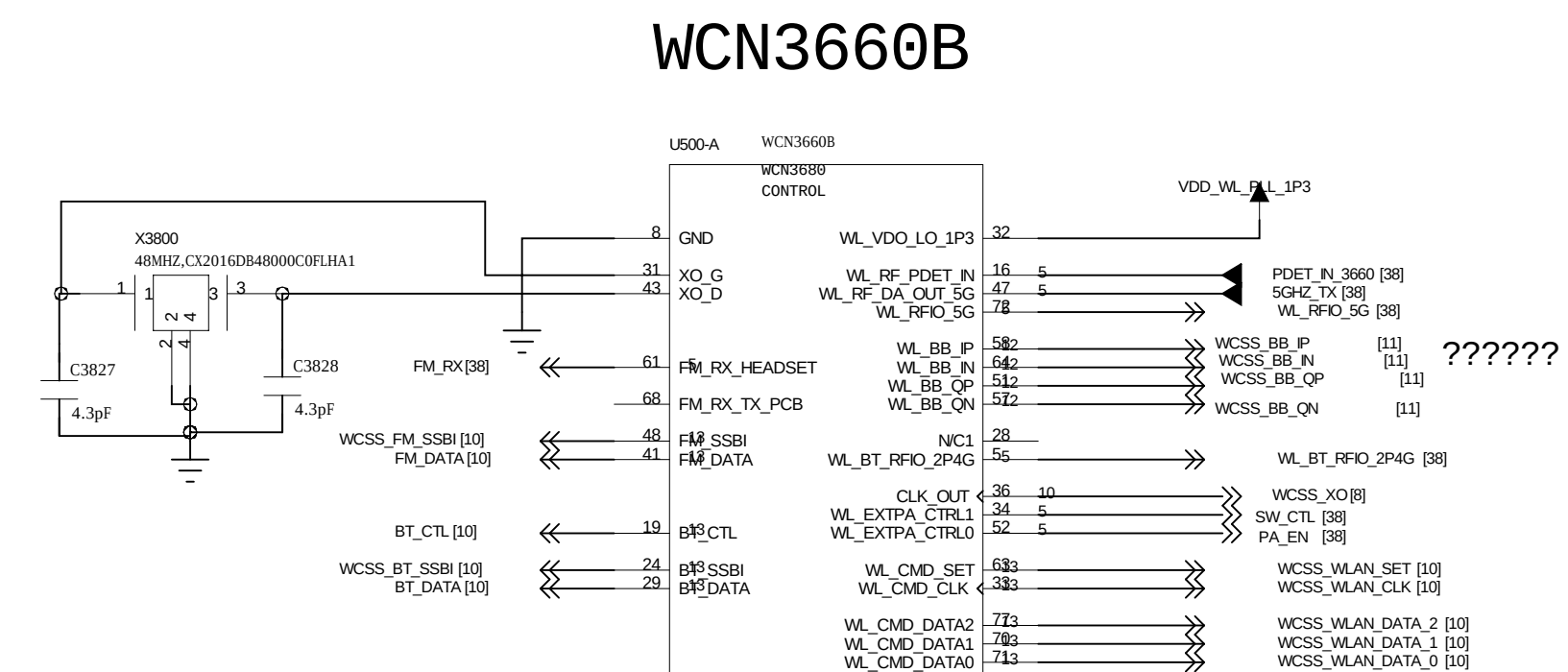
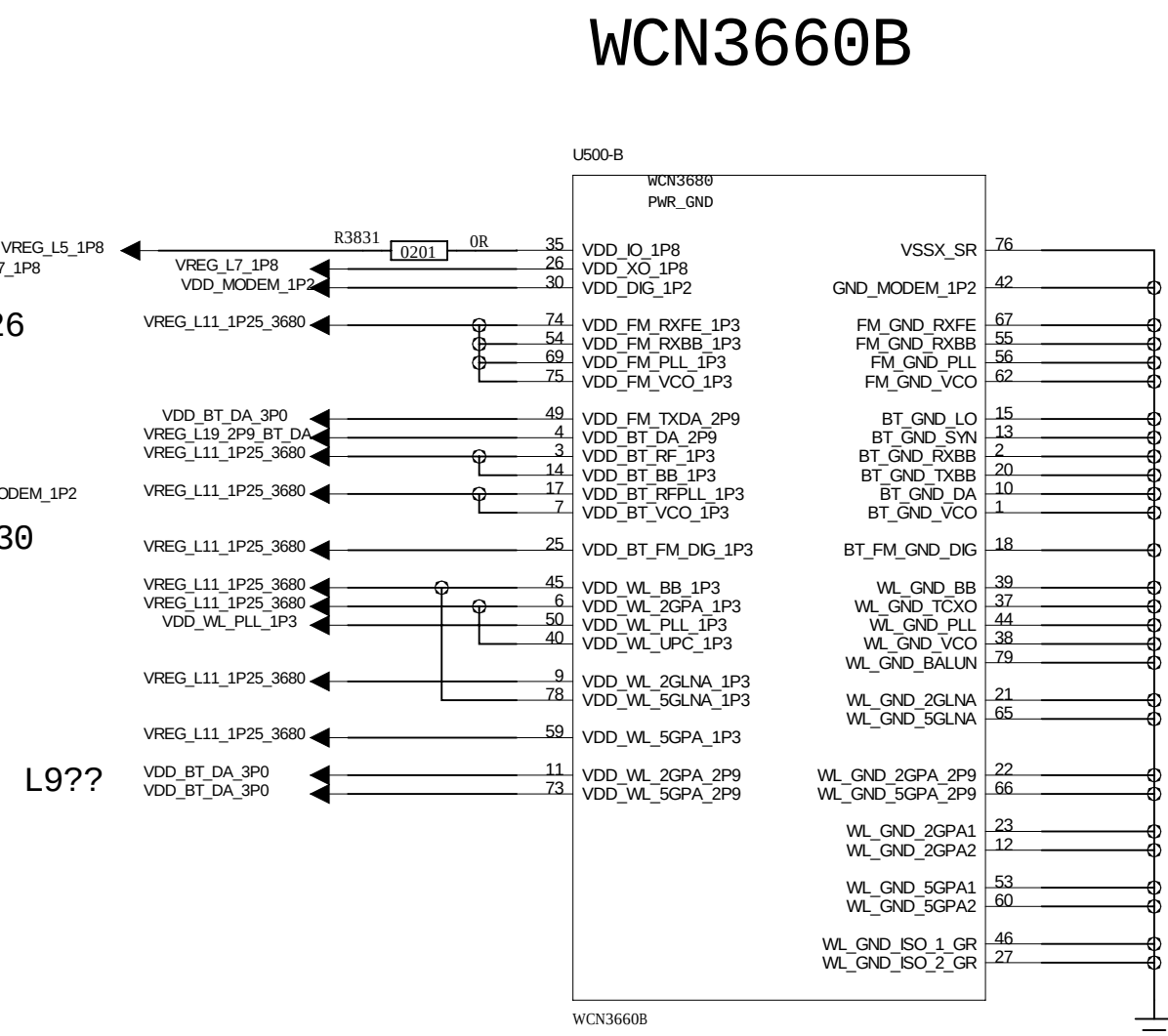
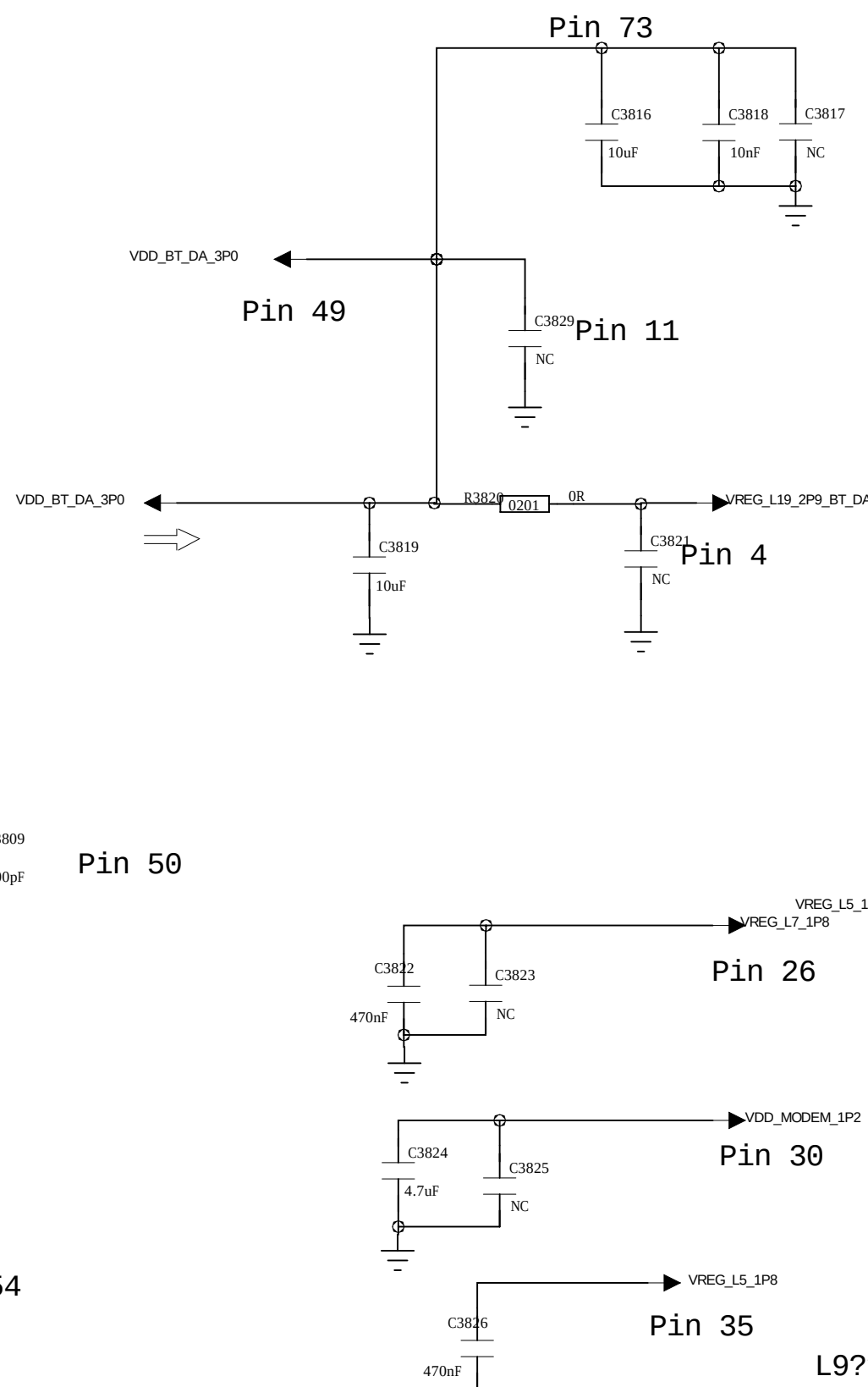
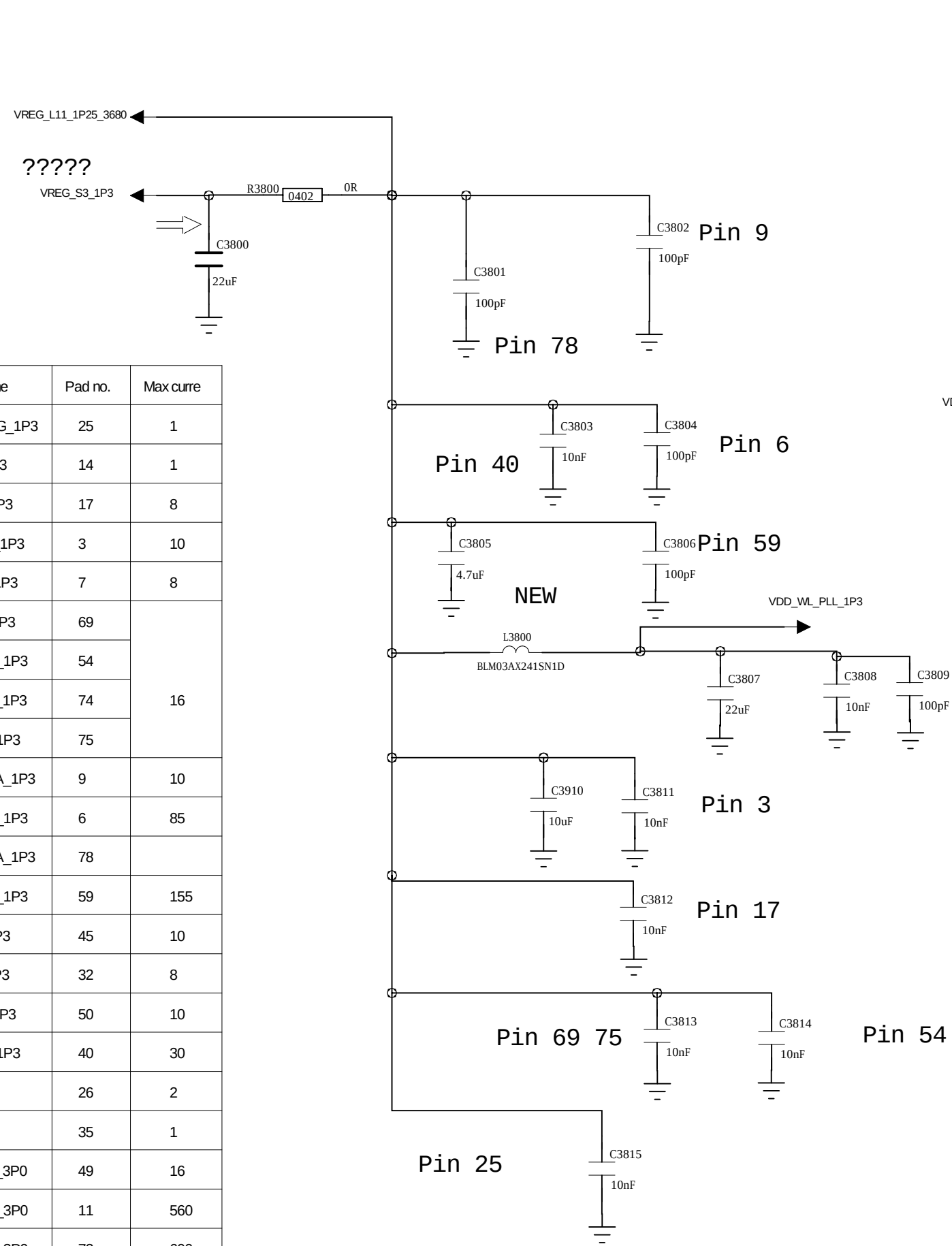
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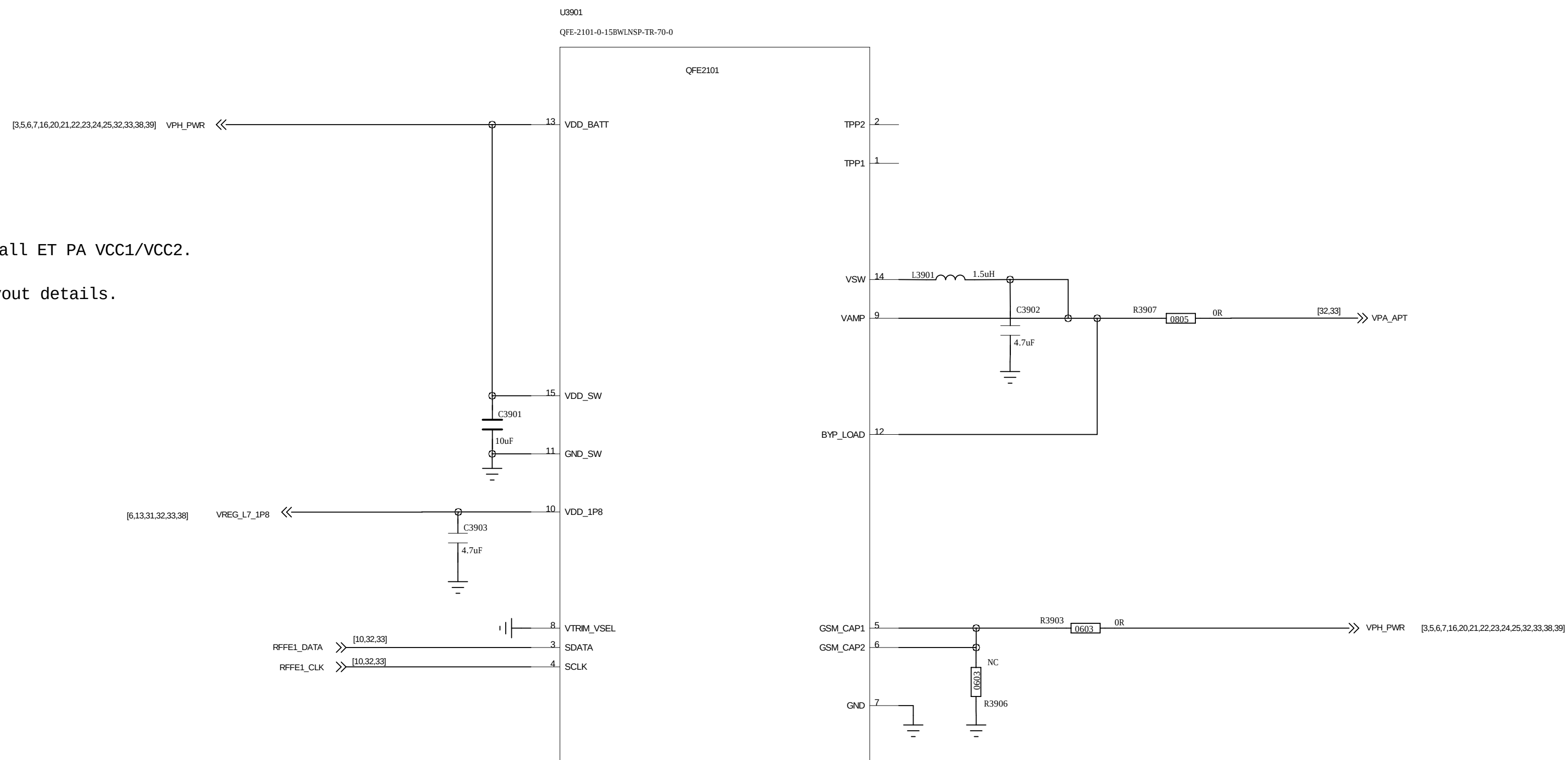
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Voltage	Pad name	Pad no.	Max current
1P3	VDD_BT_FM_DIG_1P3	25	1
	VDD_BT_BB_1P3	14	1
	VDD_BT_PLL_1P3	17	8
	VDD_BT_RXRF_1P3	3	10
	VDD_BT_VCO_1P3	7	8
	VDD_FM_PLL_1P3	69	16
	VDD_FM_RXBB_1P3	54	
	VDD_FM_RXFE_1P3	74	
	VDD_FM_VCO_1P3	75	
	VDD_WM_2GLNA_1P3	9	10
	VDD_WM_2GPA_1P3	6	85
	VDD_WM_5GLNA_1P3	78	
	VDD_WM_5GPA_1P3	59	155
	VDD_WM_BB_1P3	45	10
	VDD_WM_LO_1P3	32	8
VDD_WM_PLL_1P3	50	10	
VDD_WM_UPC_1P3	40	30	
1P8	VDD_XO_1P8	26	2
	VDD_IQ_1P8	35	1
3P0	VDD_FM_TXDA_3P0	49	16
	VDD_WM_2GPA_3P0	11	560
	VDD_WM_5GPA_3P0	73	600
	VDD_BT_TXRF_3P0	4	45
1P2	VDD_DIG_1P2	30	4



MODE	CTRL1	CTRL0
5G_RX	HIGH	LOW
5G_TX	LOW	HIGH
WLAN_OFF	LOW	LOW



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Name					
Date			Sheet	of	

0822 change list

1 | <DELETE DC block:C3214/C3219/C3418/C3412/C3414

1|<DELETE TX_B5 matching closed PA:C3274/13273/13276